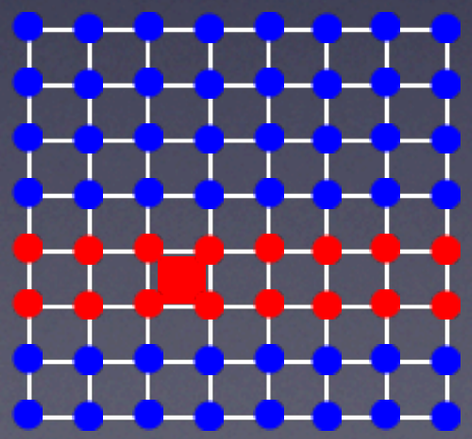
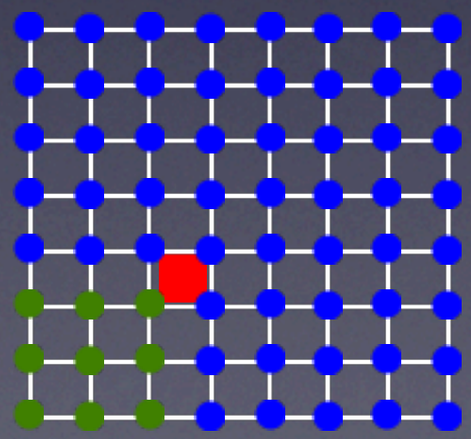
Doubling the Toric Code

Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington

0 + 1





0 + 1

Apologies

No subwoofers will be used in this talk.



Heretical*

Physics is better suited for coming up with “good” quantum codes than computer science.

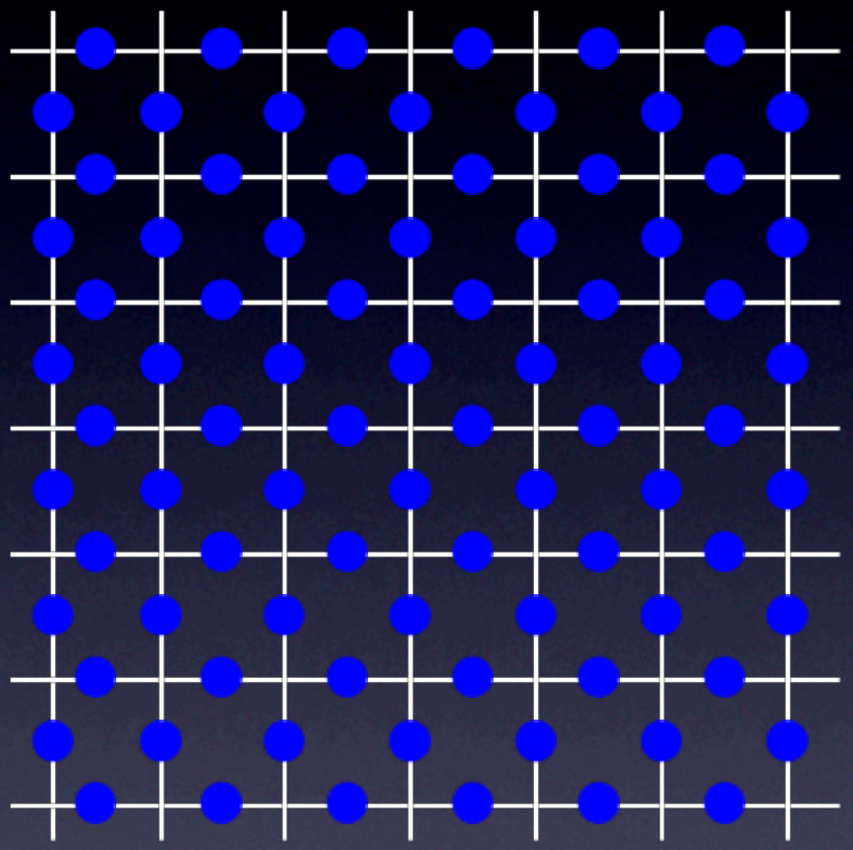
At it’s heart physics is all about **spatially local** systems interacting. This translates into codes which are better suited for architecting into a real quantum computer

* Scott Aaronson tributary slide

Today

Examples of quantum error correcting architectures which have nice spatial locality properties...

Kitaev's Toric System



periodic boundary conditions (torus)

● $= Z$ ● $= X$

plaquette operator



$$S_p = \prod_{e \in \delta p} Z_e$$

vertex operator



$$S_v = \prod_{e | v \in e} X_e$$

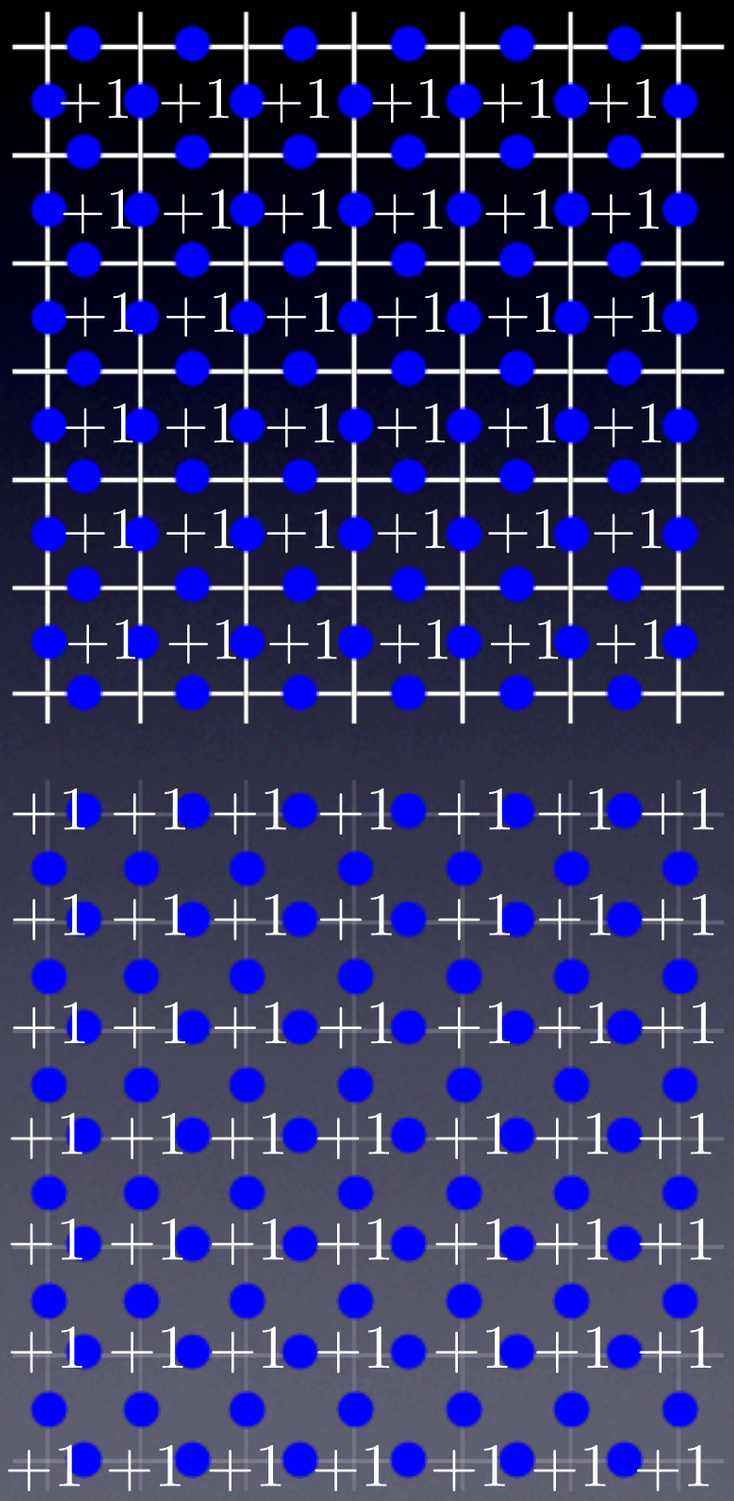
$$[S_p, S_v] = 0 \quad S_p^2 = S_v^2 = I$$

Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)

Kitaev's Toric System



$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

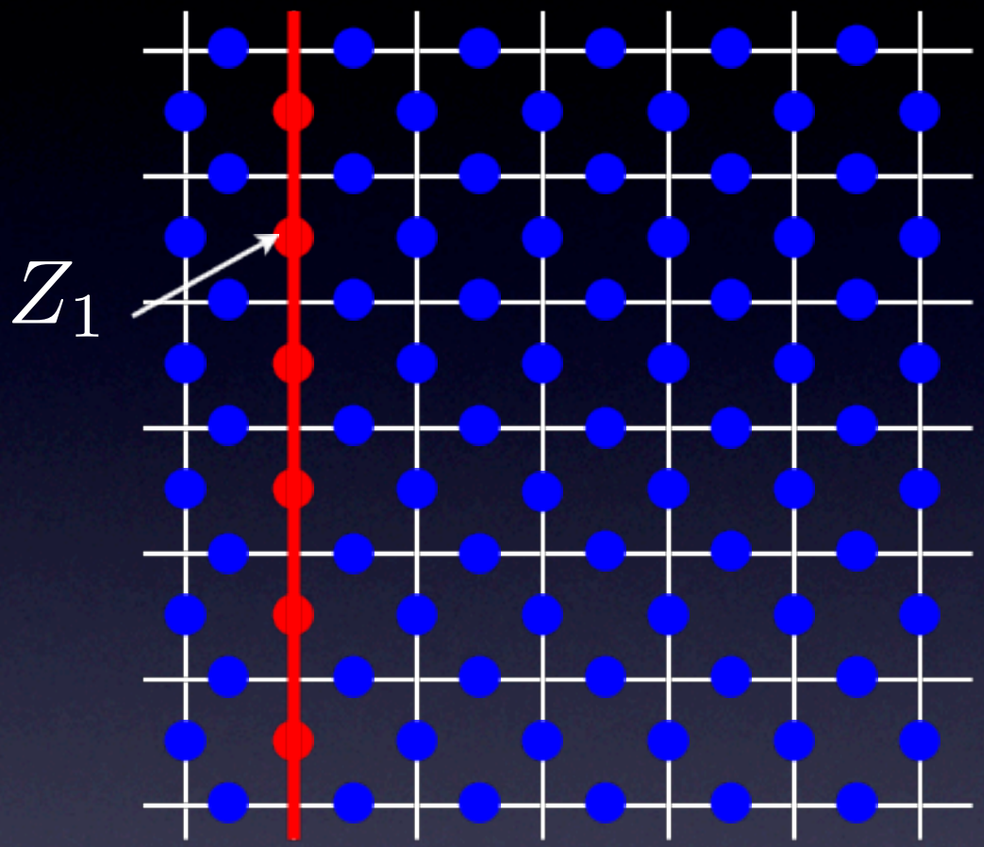
Ground state has

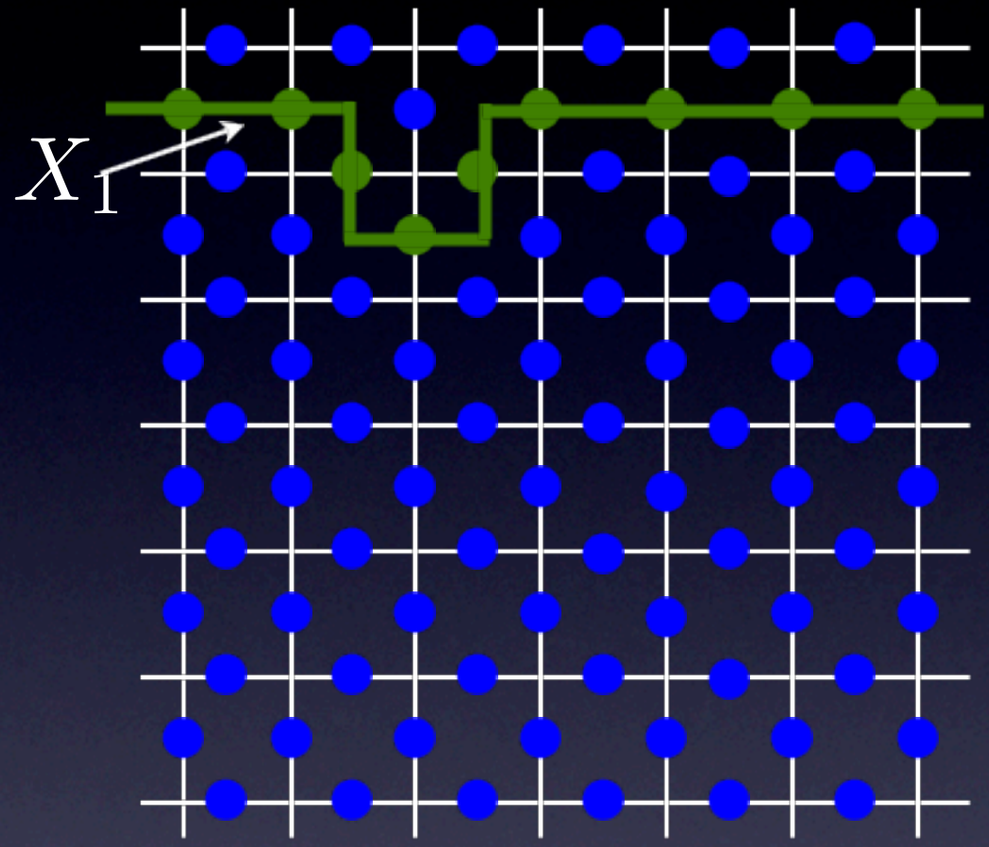
$$S_p|\psi\rangle = |\psi\rangle$$

$$S_v|\psi\rangle = |\psi\rangle$$

Degeneracy?

Kitaev's Toric Qubits



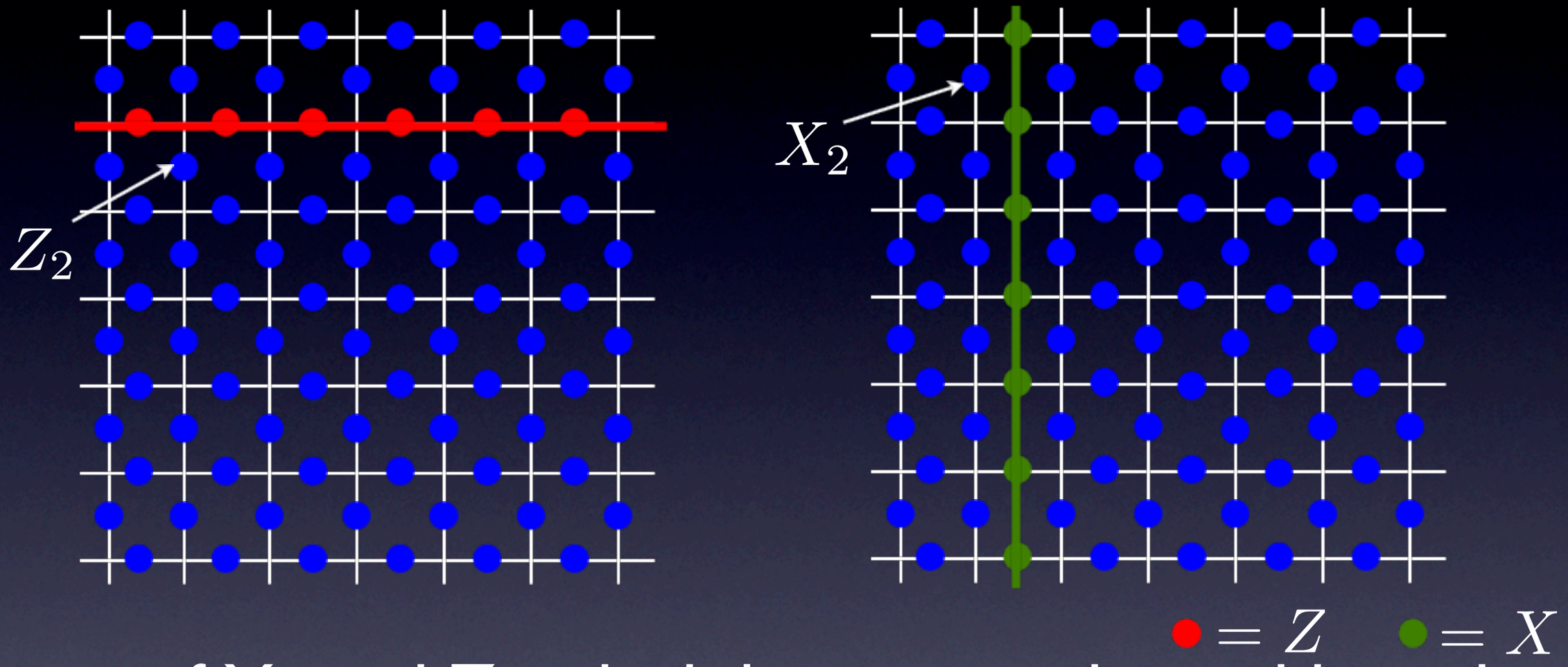


$\bullet = Z \quad \bullet = X$

strings of Xs and Zs which loop around an odd number of times commute with S_p, S_v but cannot be generated as products of these operators

$$[S_p, X_1] = [S_v, X_1] = [S_p, Z_1] = [S_v, Z_1] = 0 \quad [X_1, Z_1] \neq 0$$

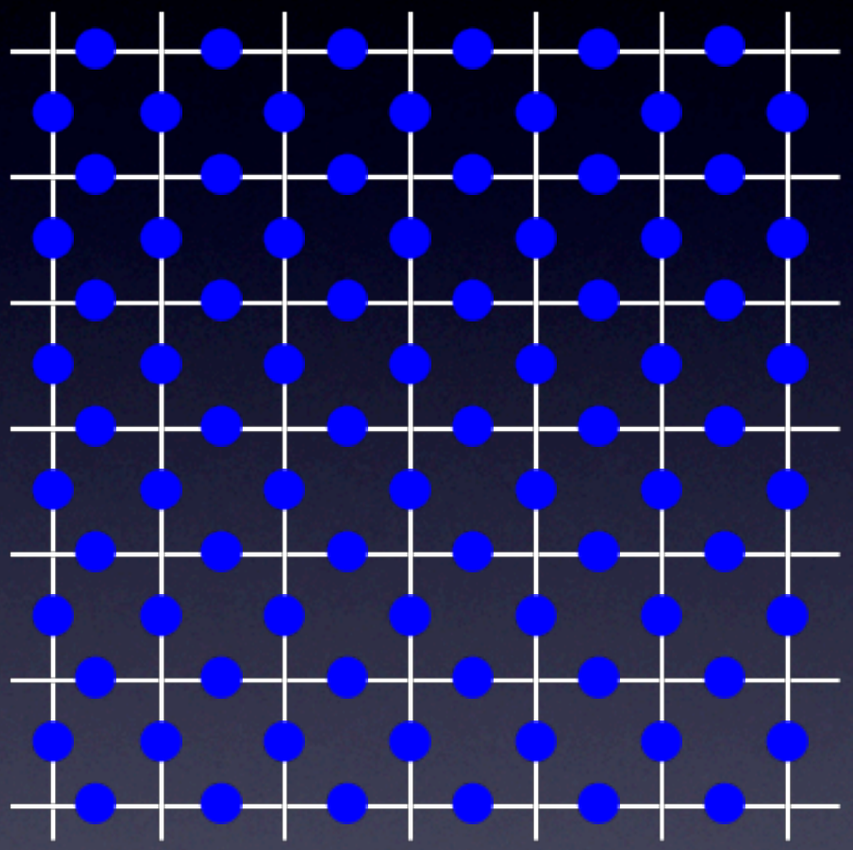
Kitaev's Toric Qubits



strings of Xs and Zs which loop around an odd number of times commute with S_p, S_v but cannot be generated as products of these operators

$$[S_p, X_2] = [S_v, X_2] = [S_p, Z_2] = [S_v, Z_2] = 0 \;\; [X_2, Z_2] \neq 0$$

Kitaev's Toric System



Kitaev Hamiltonian:

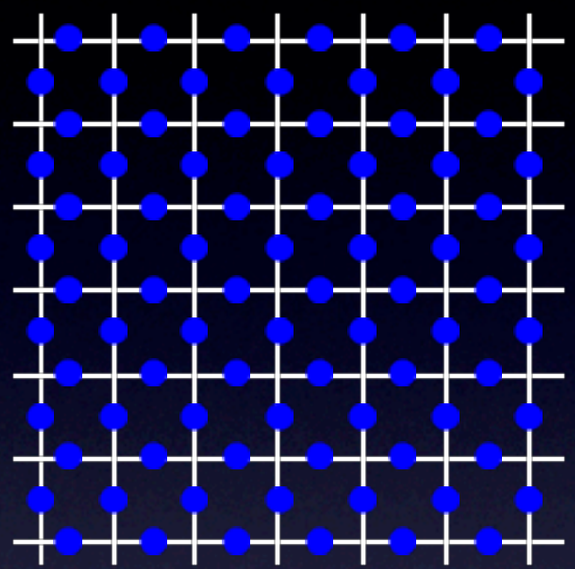
$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$



ground state is a quantum error correcting code

$2L^2$ qubits, encodes 2 qubits, can correct L/2 qubit errors

Kitaev's Toric Protection



Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

$$\frac{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}{2\Delta}$$

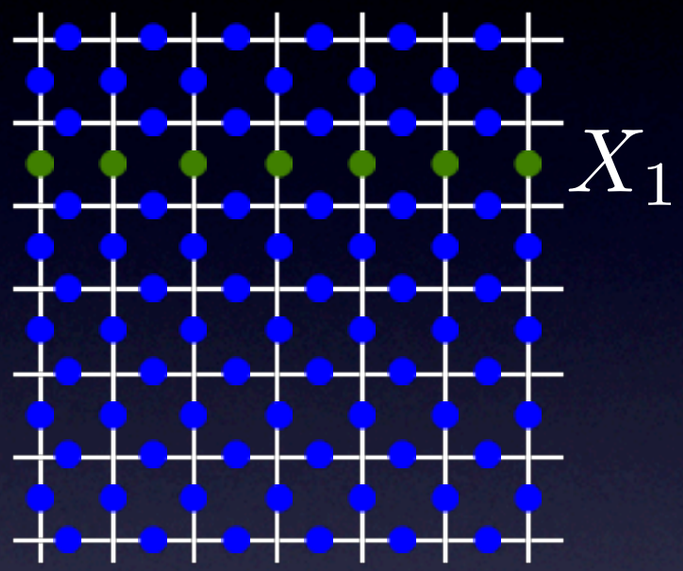
Single qubit perturbation $V = \lambda \sum_e P_e$

$$\delta E = \frac{\lambda^L}{\Delta^{L-1}}$$

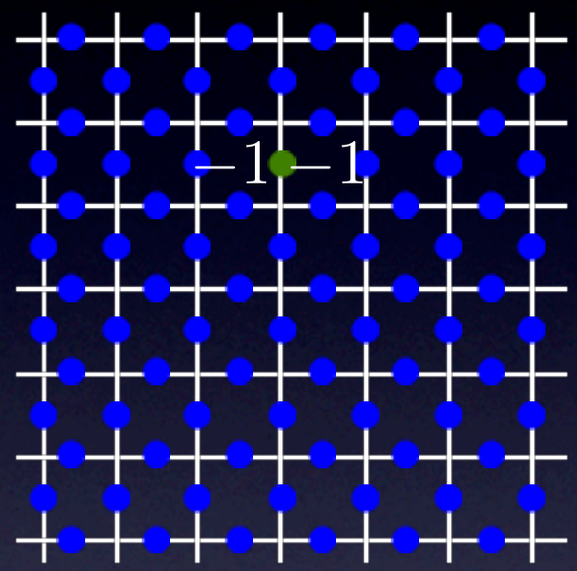
Exponential suppression of level splitting due to external (local) perturbations...but disorders at finite temperature

Kitaev's Toric Problem

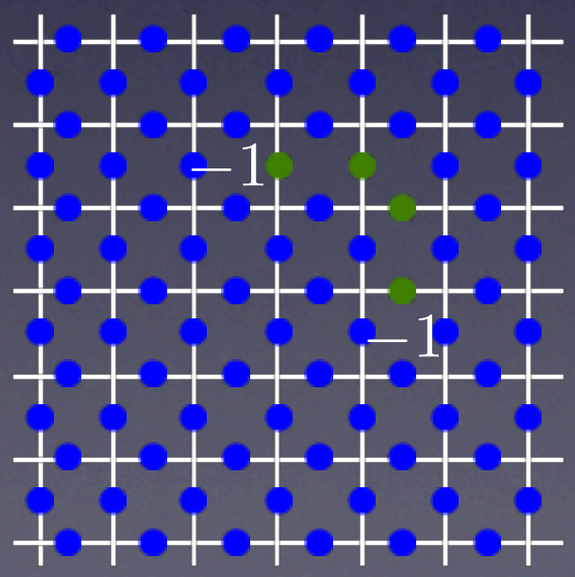
in implementing a gate



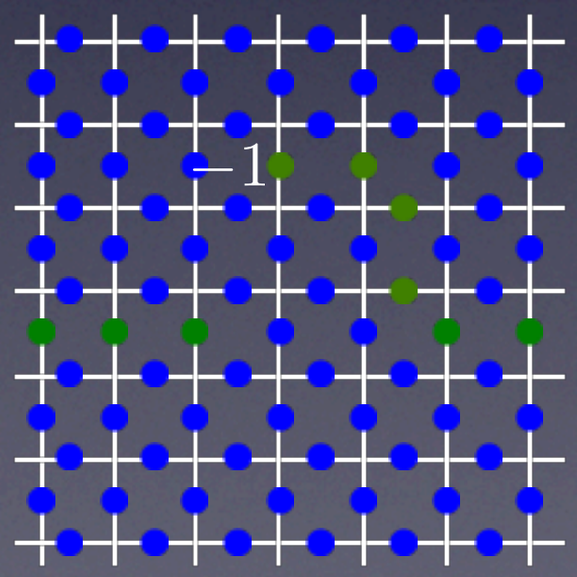
we may produce an error



which is a free particle



& turns into uncorrectable error



$\bullet = Z$

$\bullet = X$

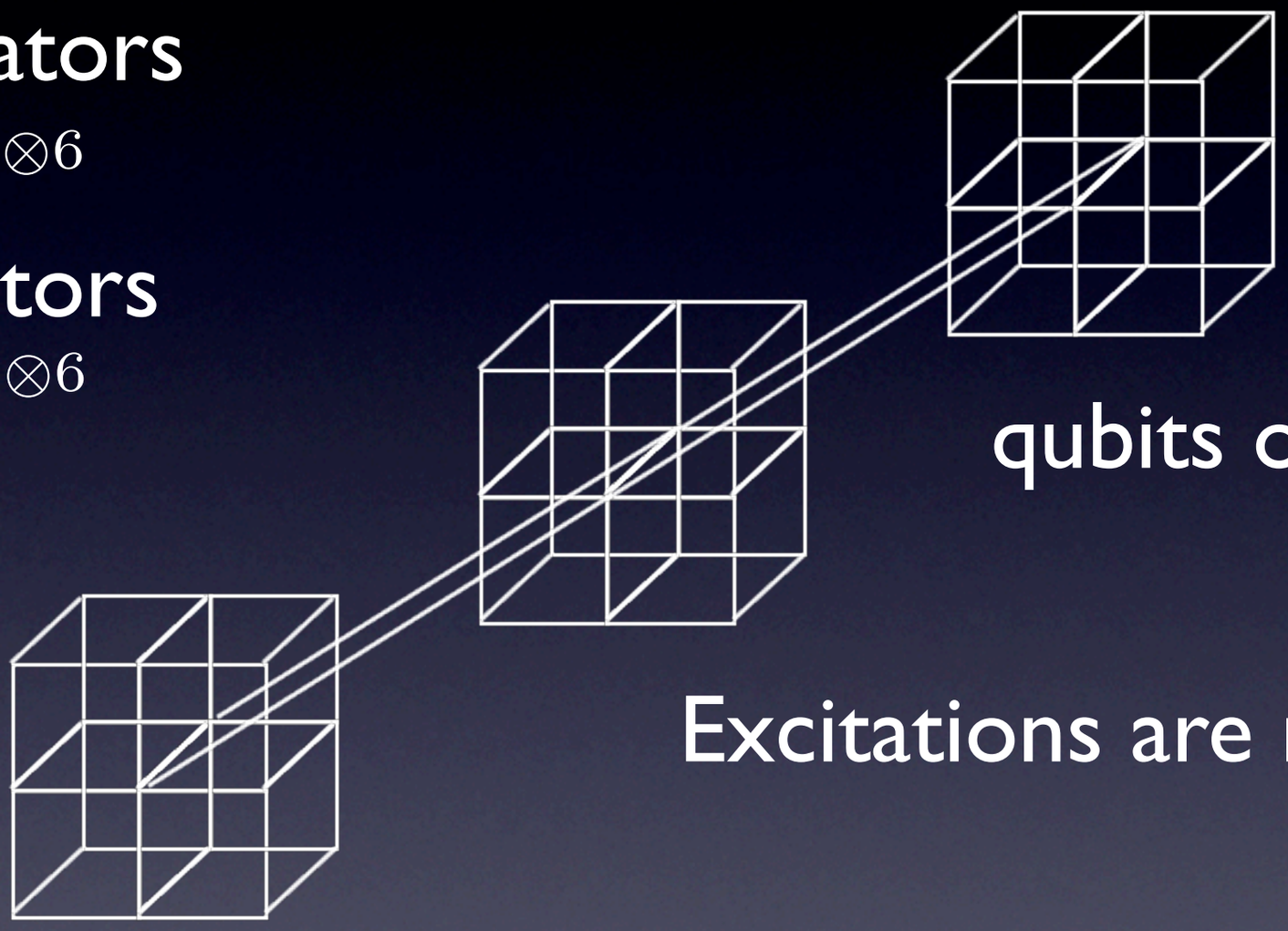
4D Toric Code

cube operators

$$S_c = Z^{\otimes 6}$$

link operators

$$S_l = X^{\otimes 6}$$



qubits on plaquettes

Excitations are now strings

4D code is **self-correcting**

Dennis, Kitaev, Landahl, Preskill "Topological quantum memory" (2002)



Challenge: Does there exist a self-correcting quantum system in 3 spatial dimensions?

and in the meantime....the computer engineers...

Stabilizer Codes

Pauli group on n qubits: $\langle i^k P_1 \otimes P_2 \otimes \cdots P_n \rangle$

$P_i \in \{I, X, Y, Z\}$

Stabilizer group: Abelian subgroup of Pauli group not containing $-I$

$$\mathcal{S} = \langle S_1, S_2, \ldots, S_{n-k} \rangle$$

Stabilizer code: subspace fixed by stabilizer group

$$S_i |\psi_C\rangle = |\psi_C\rangle$$

Normalizer of stabilizer in Pauli group: logical operators

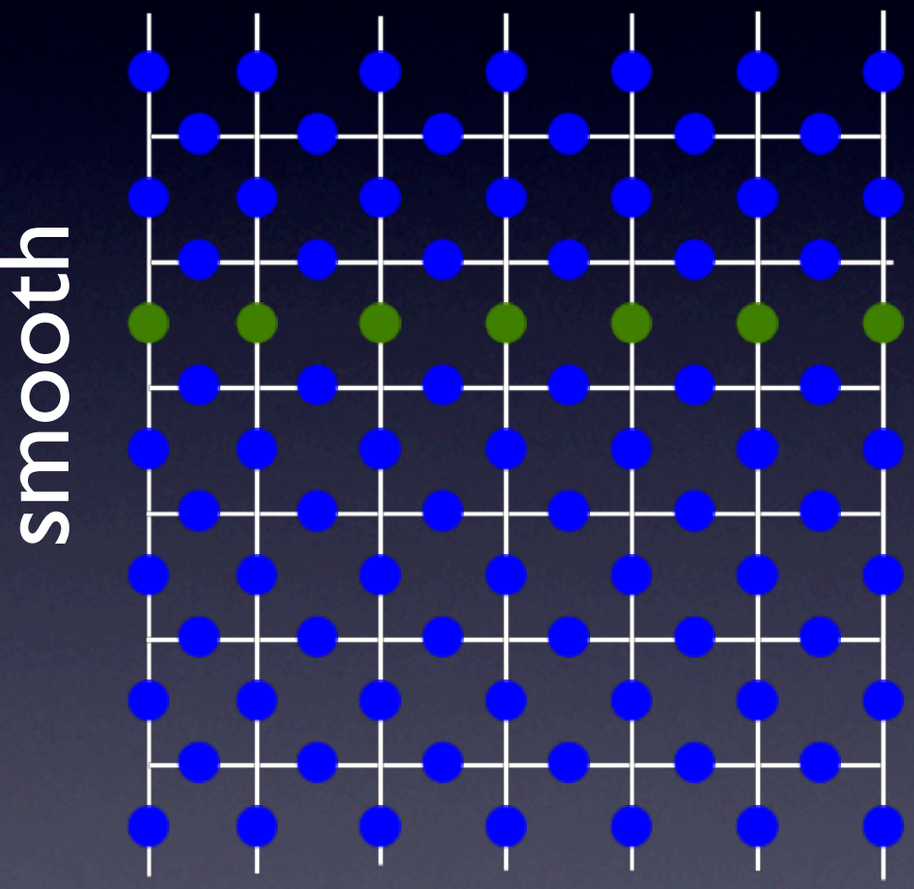
$$\mathcal{N}/\mathcal{S} \quad \begin{matrix} X_1 & X_2 & \cdots & X_k \\ Y_1 & Y_2 & \cdots & Y_k \\ Z_1 & Z_2 & \cdots & Z_k \end{matrix}$$

[[n,k,d]] code

d: smallest weight error on code

Surface Code

rough



smooth

plaquette operator



$$S_p = \prod_{e \in \delta p} Z_e$$

vertex operator



$$S_v = \prod_{e | v \in e} X_e$$

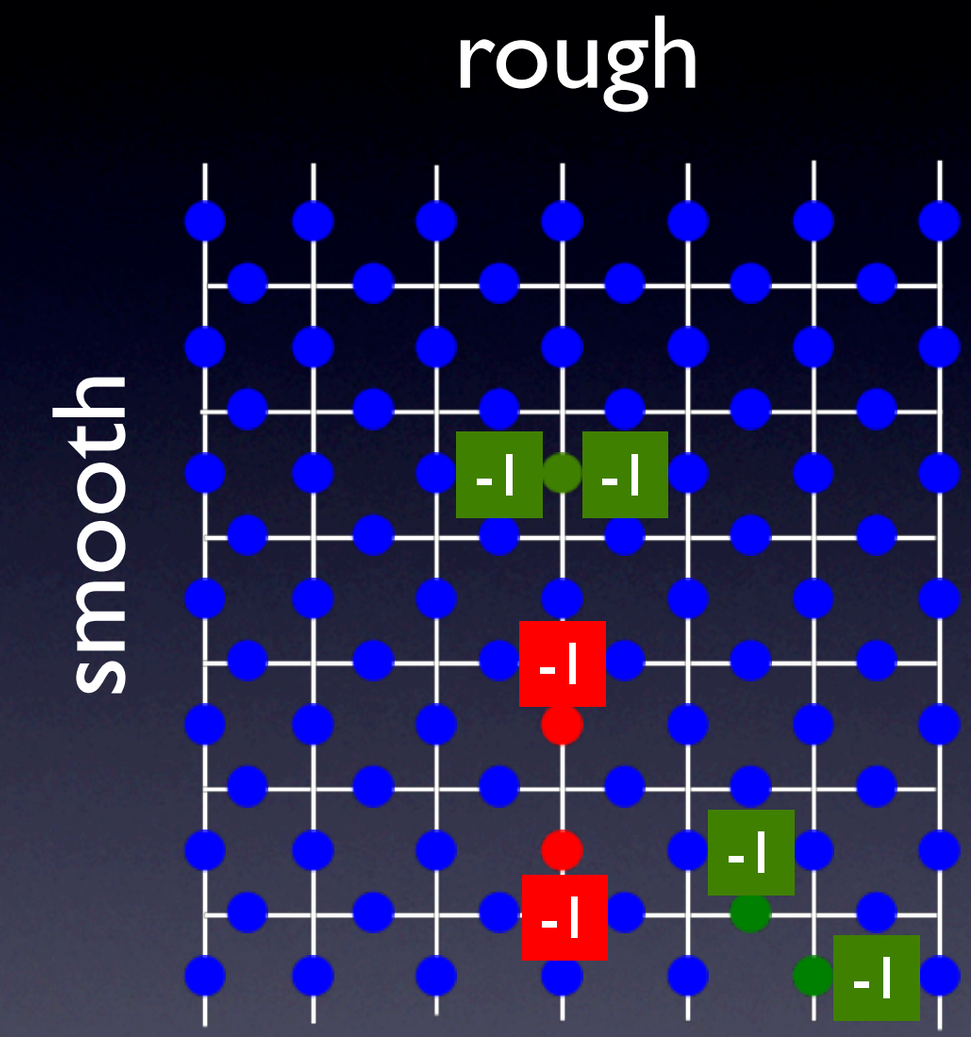
encodes one qubit

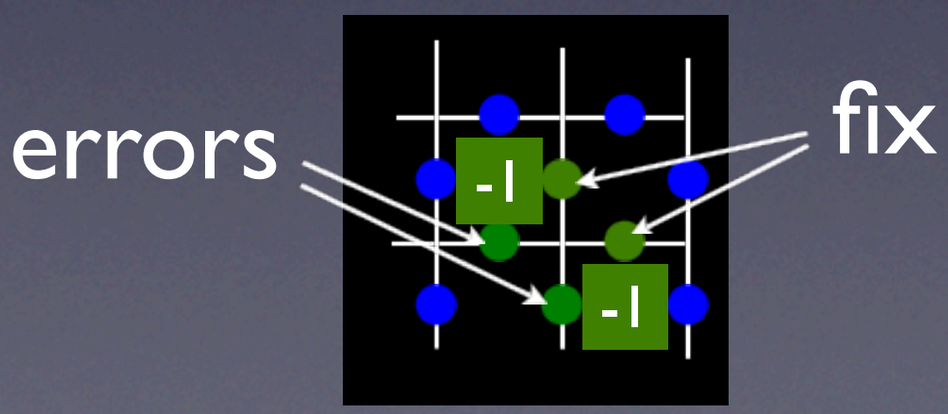
$\mathcal{N}/\mathcal{S}$ $Z_L$string of Z's linking two smooth boundaries

$X_L$string of X's linking two smooth boundaries

Dennis, Kitaev, Landahl, Preskill "Topological quantum memory" (2002)

Surface Code





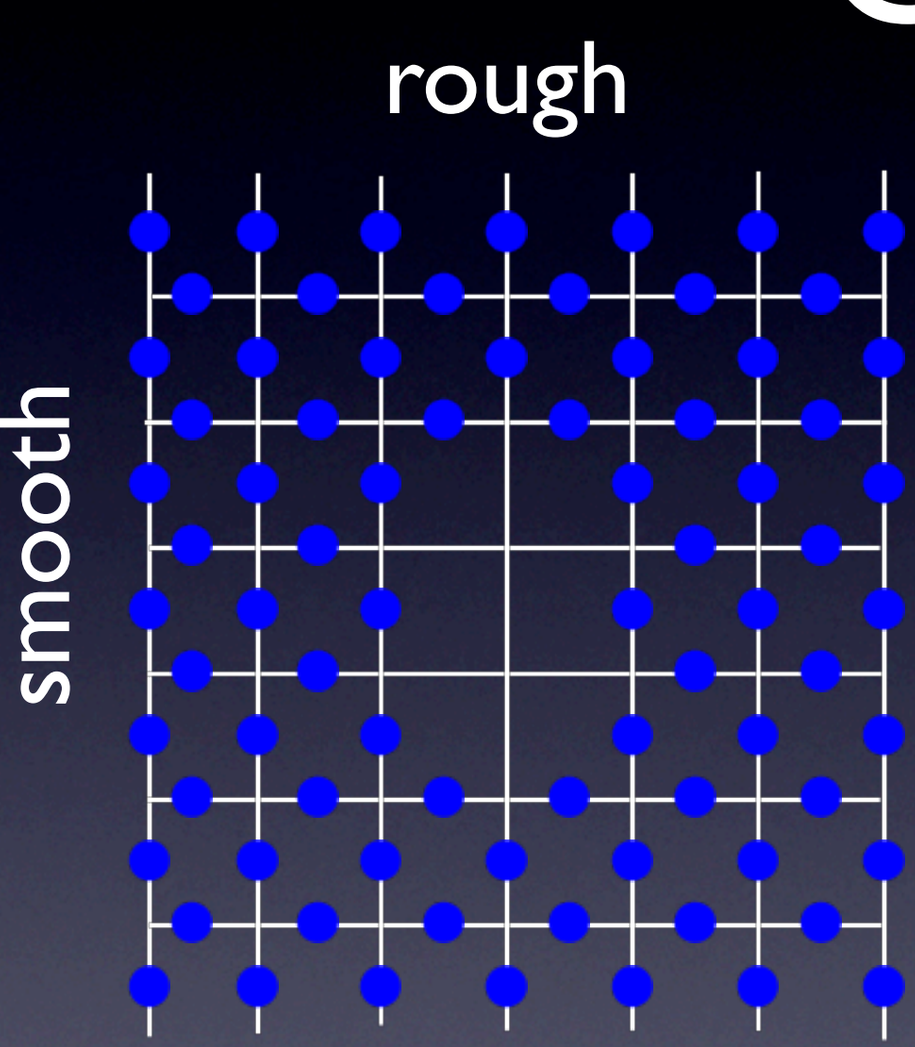
Error correction procedure:

1. Measure plaquette and vertex operators

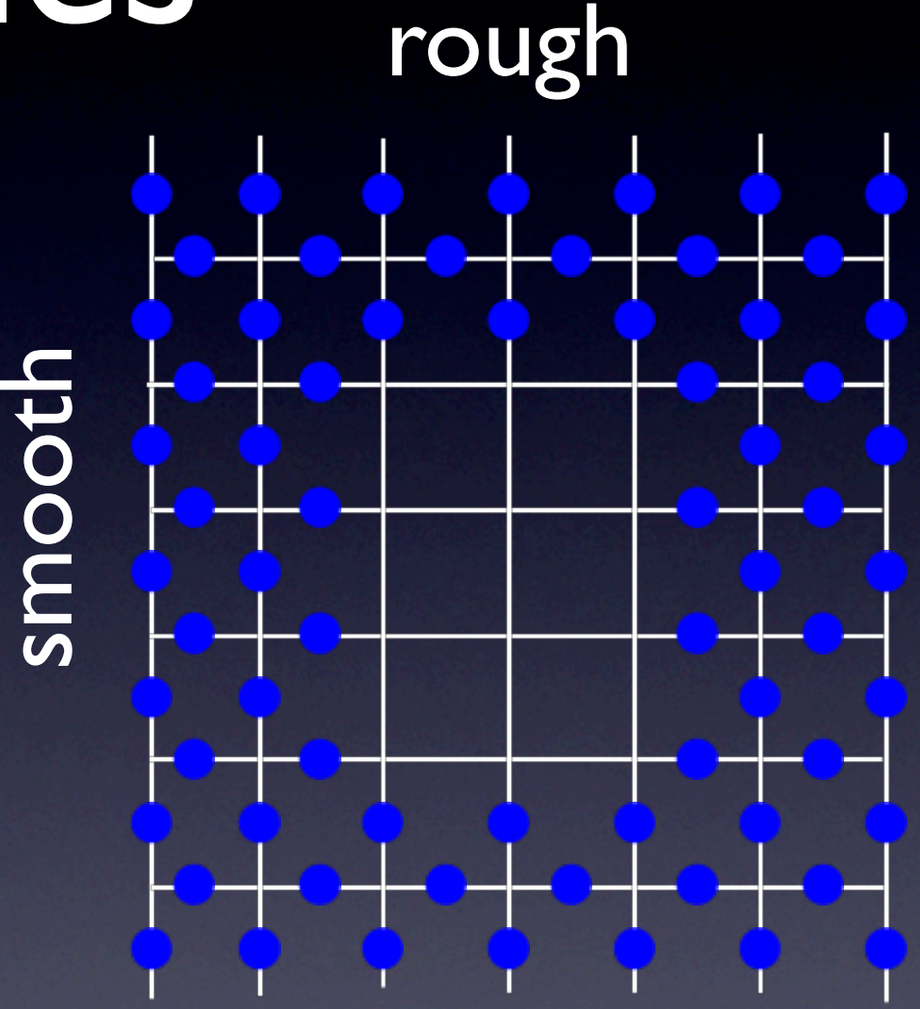
2. graph G_X/G_Z with Z/X -1 syndromes as nodes anddistance between errors as weights

3. Find perfect minimal weighted matching & apply corrective chains between pairs

Puncturing Surface Codes

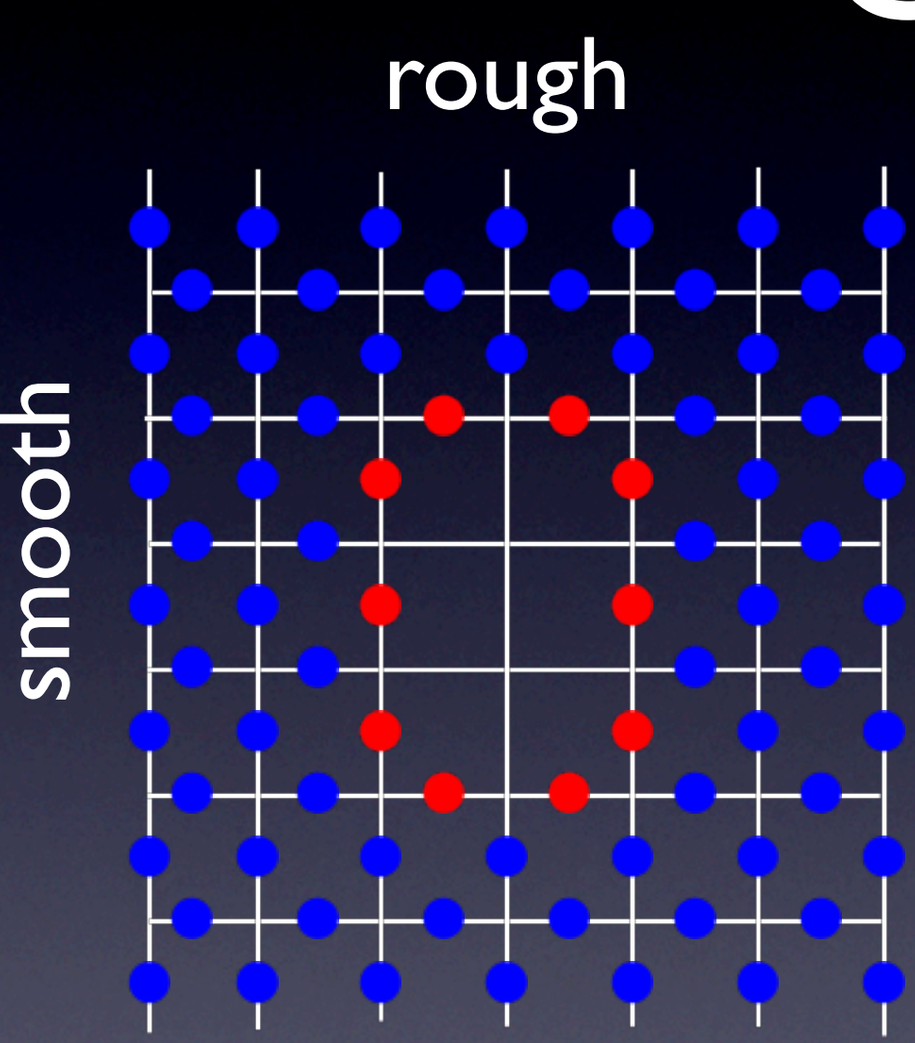


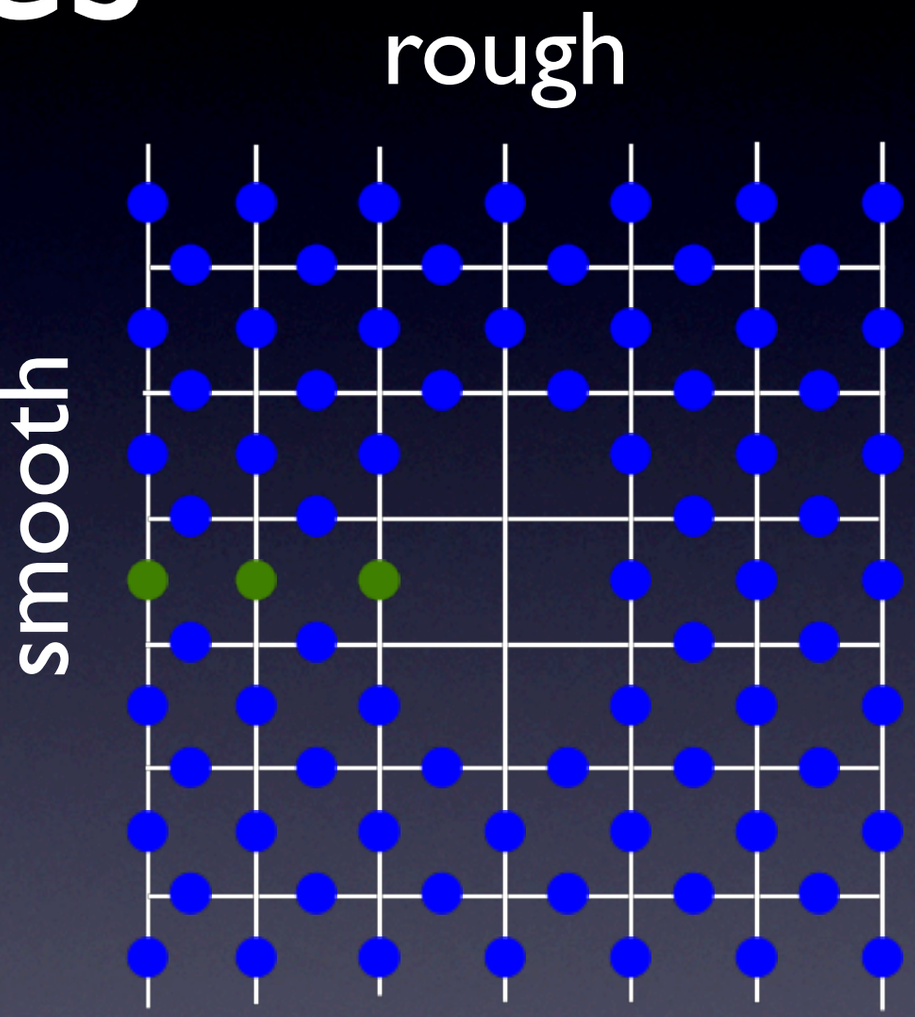
smooth defect



rough defect

Puncturing Surface Codes

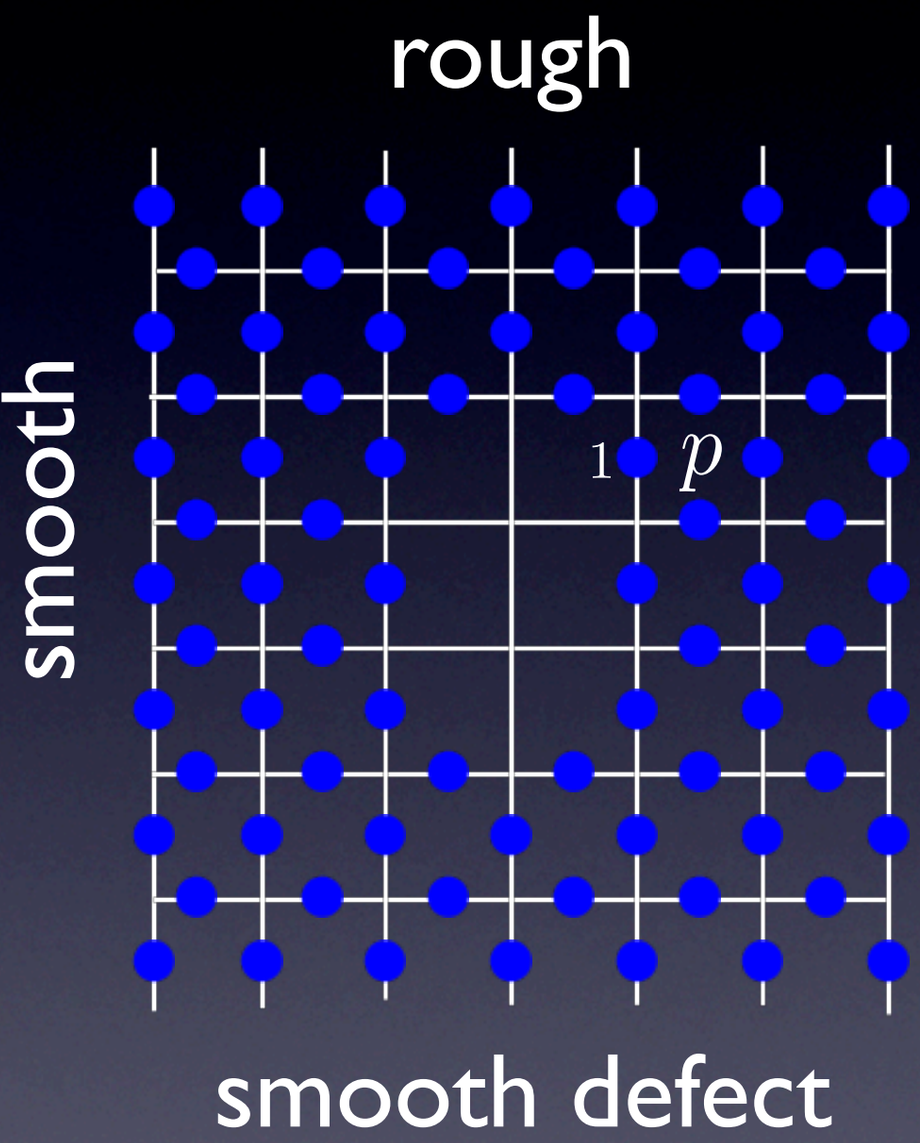




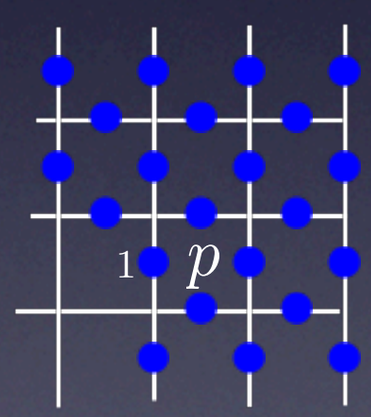
Z_L smooth defect X_L

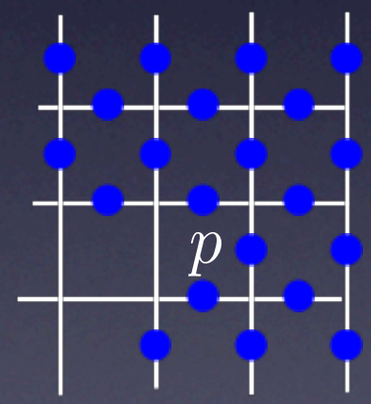
distance of new qubit is minimum of length to smooth wall / perimeter of defect

Moving Defects



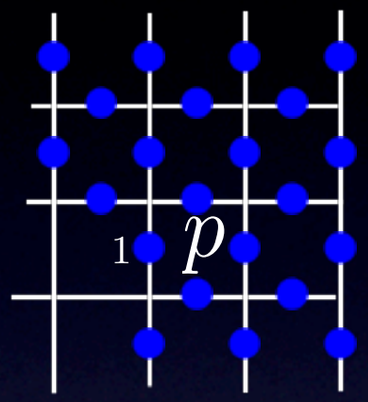
1. Measure qubit 1 in X basis
2. Conditional on outcome -1 apply plaquette p operator to fix vertex stabilizers

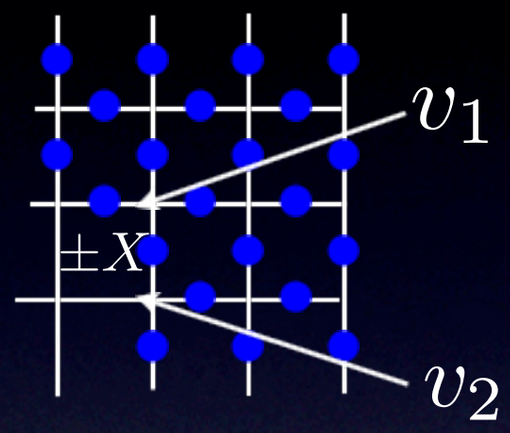


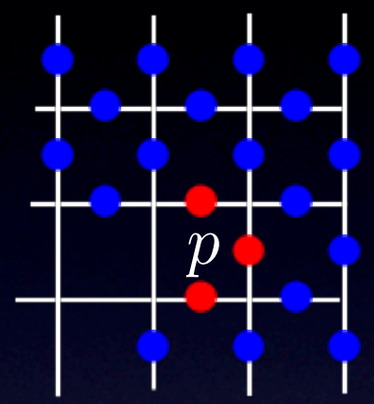


Fowler, Stephens, and Groszkowskid, "High threshold universal quantum computation on the surface code" (2008)

Moving Defects







$$S_p \to \pm X$$

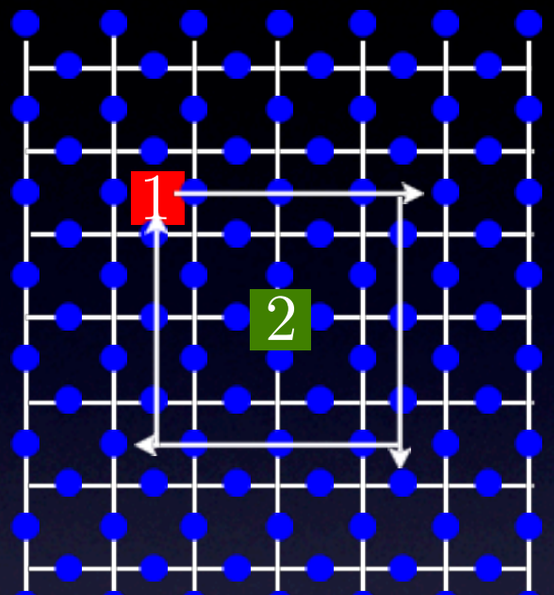
$$S_{v_1} \to \pm X^{\otimes 3}$$

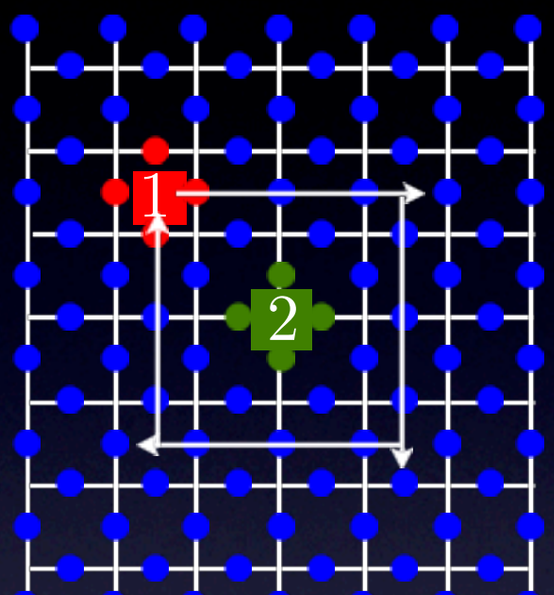
$$S_{v_2} \to \pm X^{\otimes 2}$$

Similar procedure (measure stabilizer conditionally apply single X) to move wall inward

Fowler, Stephens, and Groszkowskid, "High threshold universal quantum computation on the surface code" (2008)

Braiding Defects





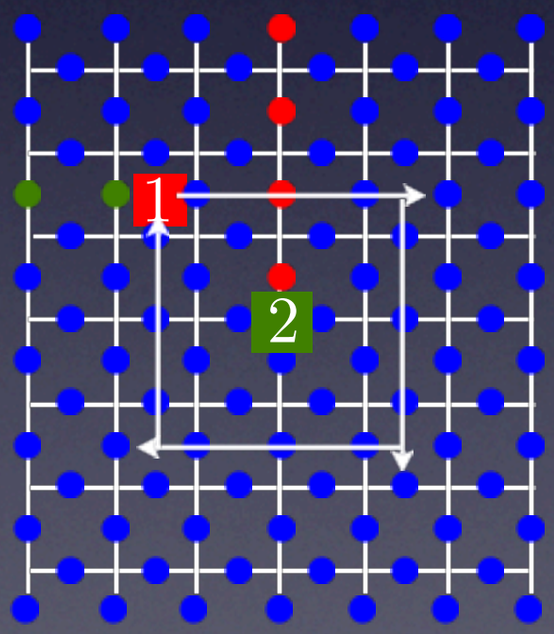
$$Z_1 \to Z_1$$

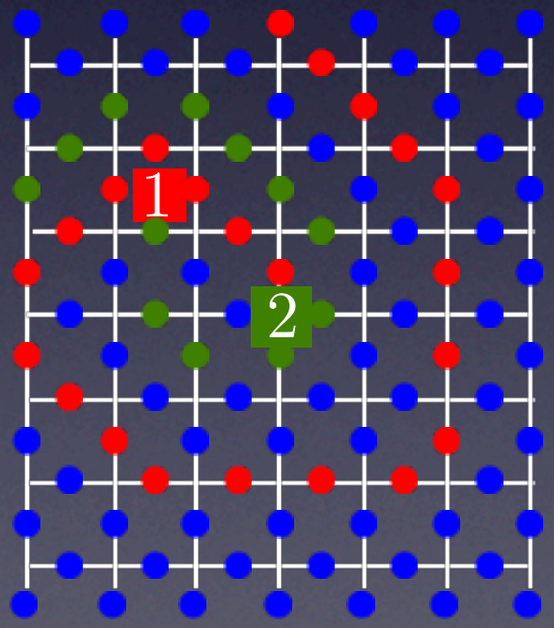
$$X_2 \to X_2$$

$$X_1 \to X_1 X_2$$

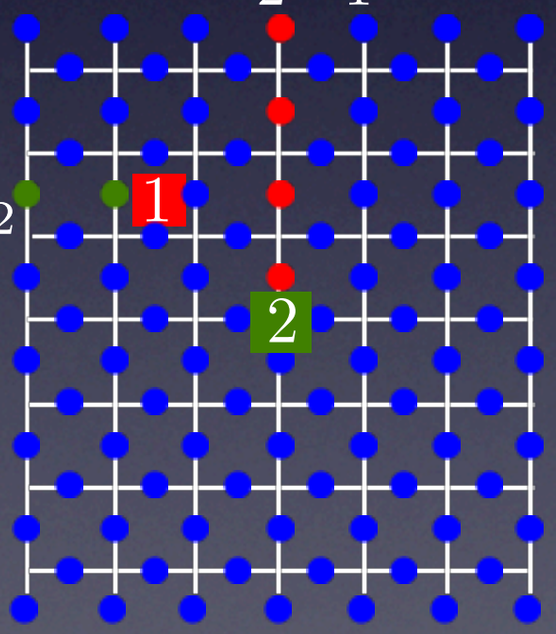
$$Z_2 \to Z_1 Z_2$$

$Z_2 Z_1$





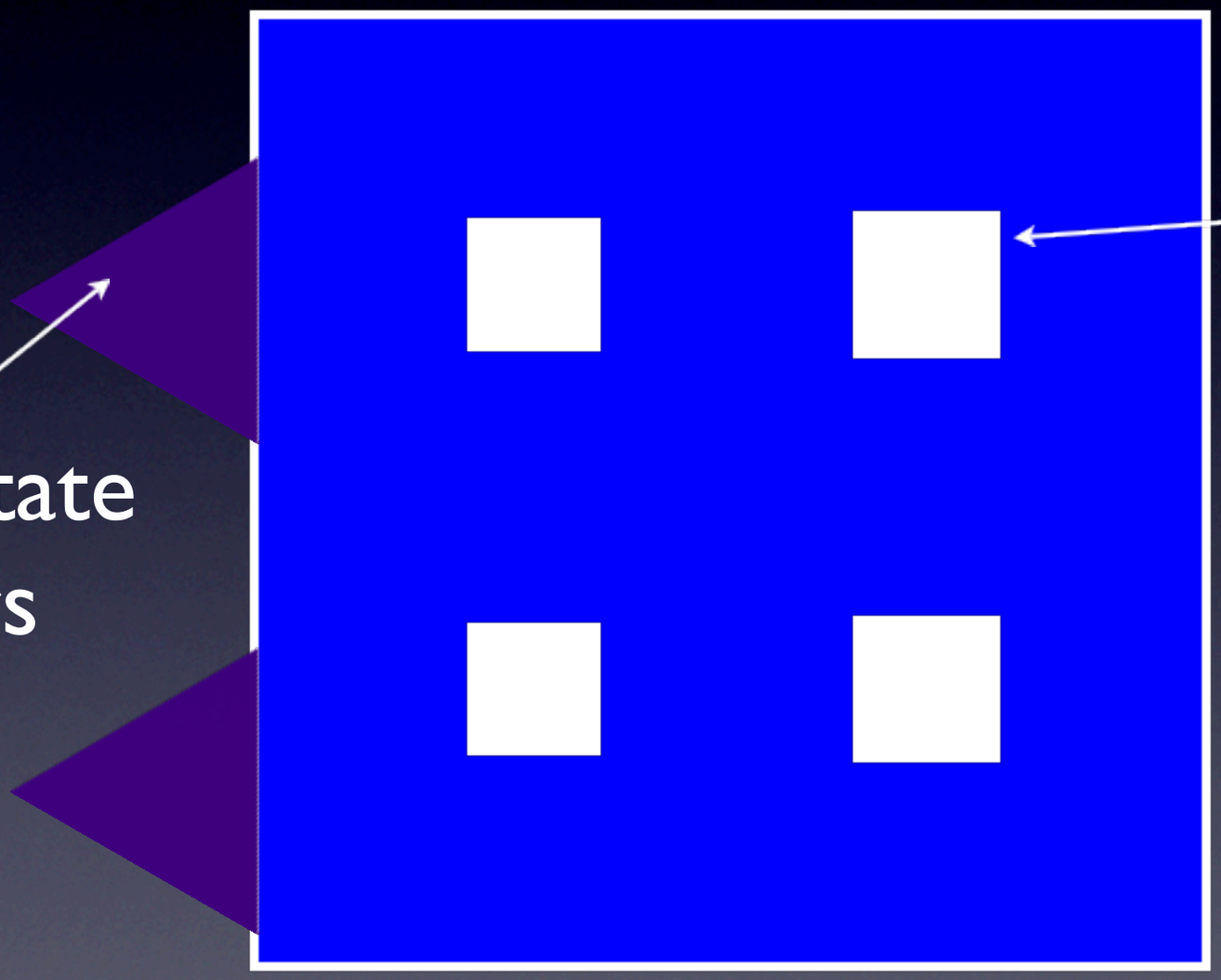
$X_1 X_2$



Controlled Not

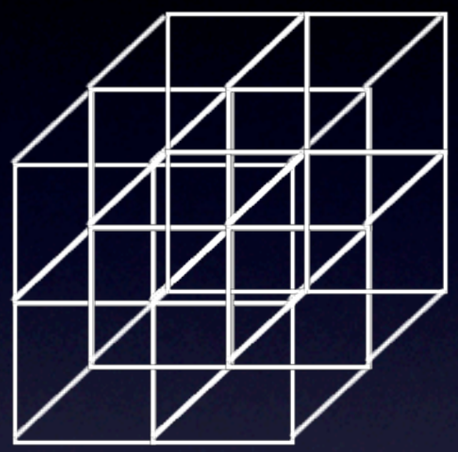
Local Architecture

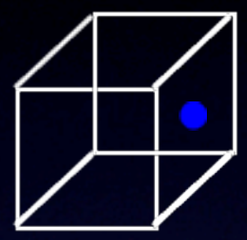
Build a nearly local quantum error correcting architecture:



Fowler, Stephens, and Groszkowskid, "High threshold universal quantum computation on the surface code" (2008)

3D Toric Code



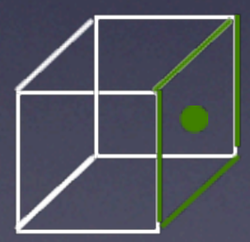


qubits on faces

link operators: $S_l = \prod_{l \in \delta p} Z_p$

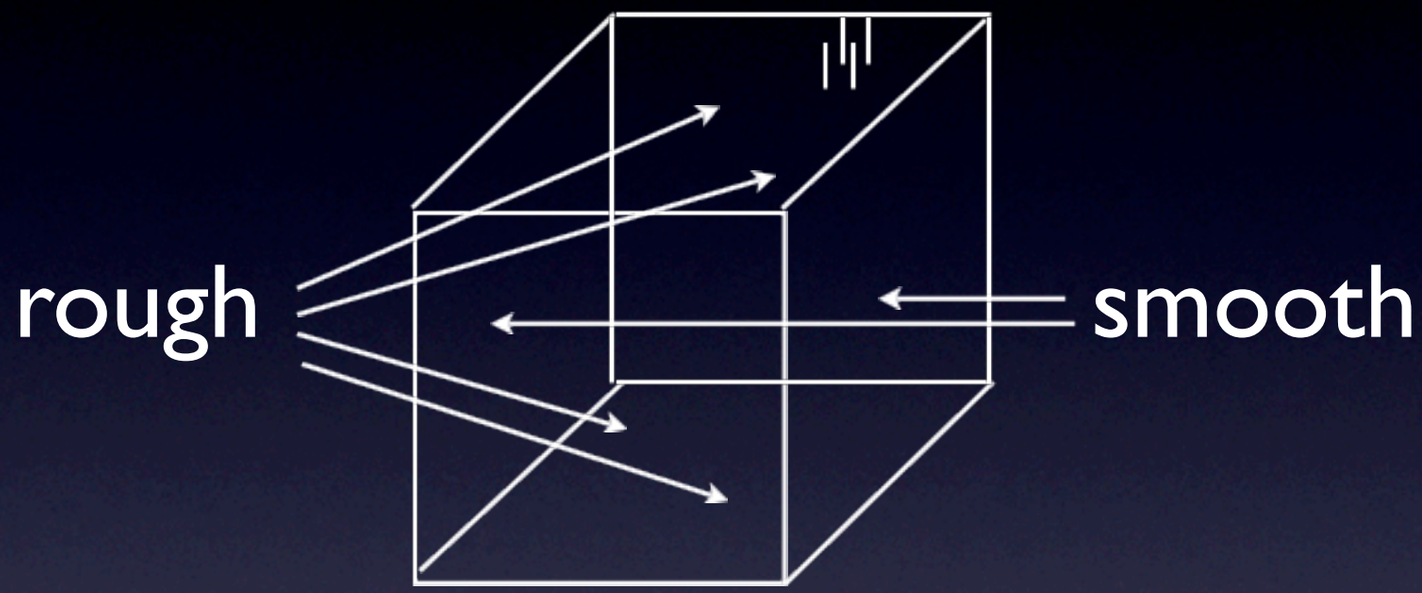
cube operators: $S_c = \prod_{p \in \delta c} X_p$

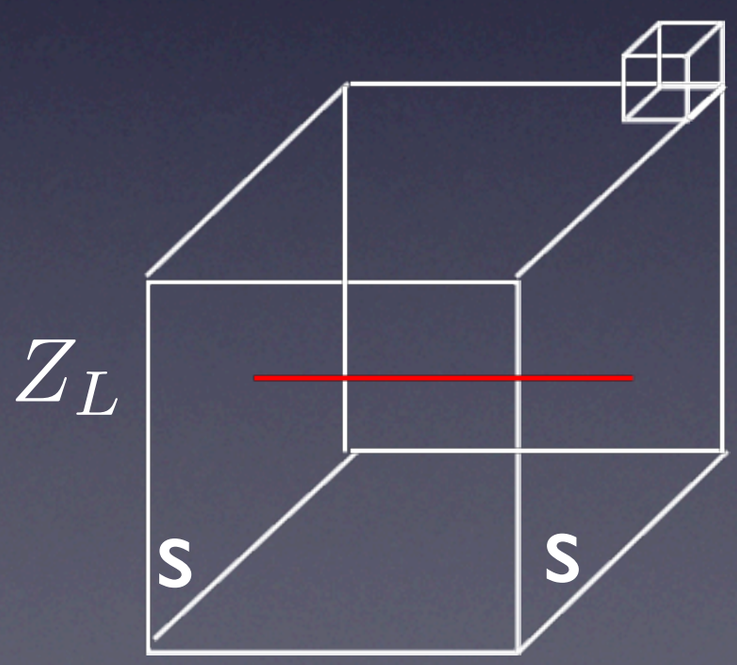
X errors are “strings”
Z errors are “particles”

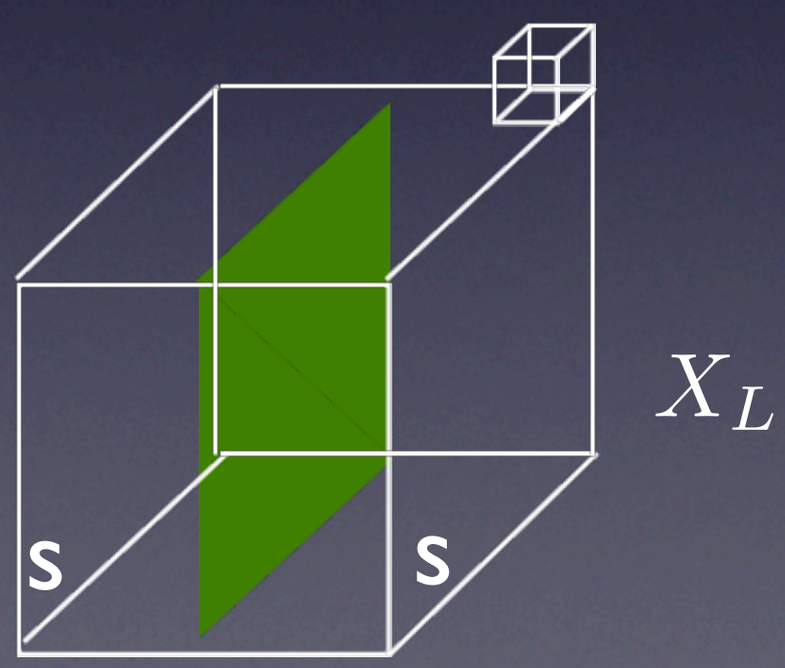


In three dimensions how to get nontrivial statistics?

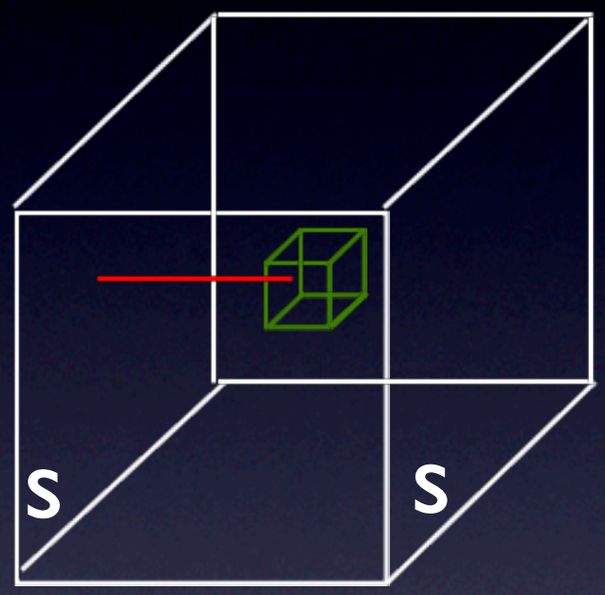
3D Toric Code







3D Toric Code Defects

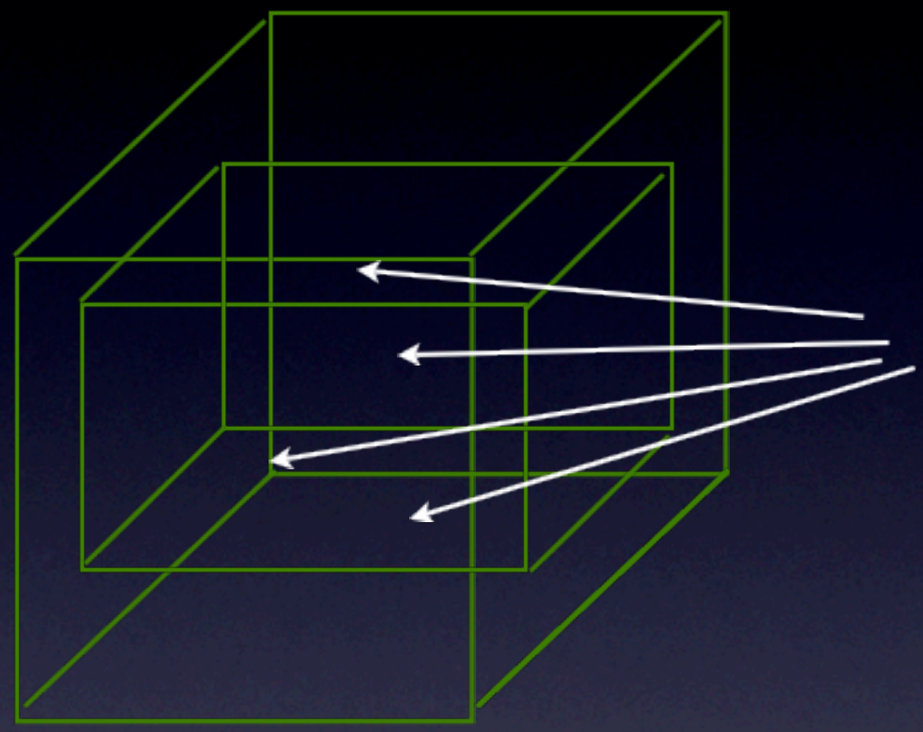


cube defects

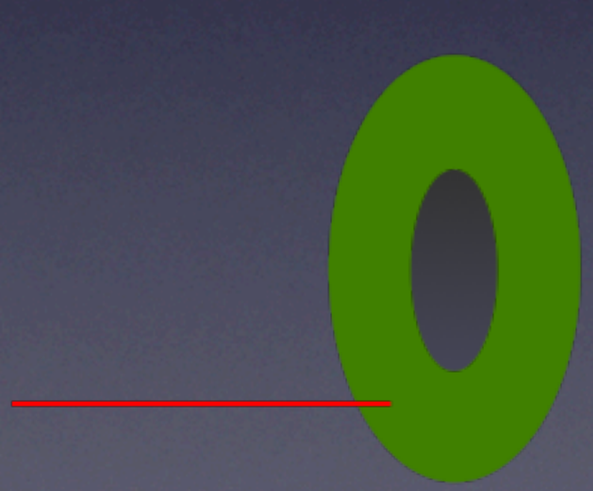
but braiding these around each other can always be undone using stabilizer operators....

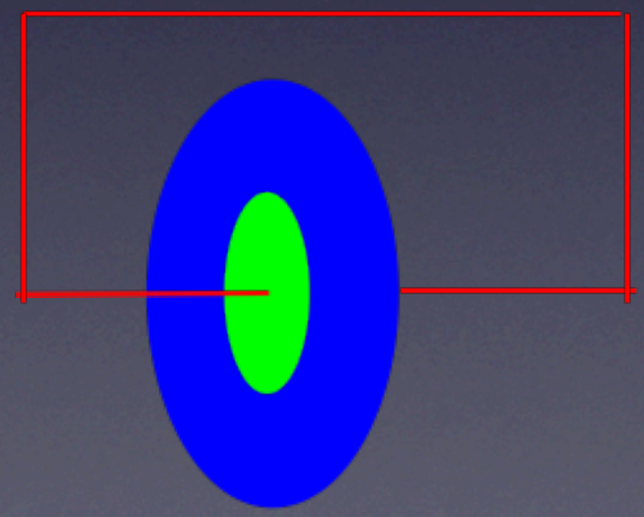
3D Toric Code Defects

cut out a torus



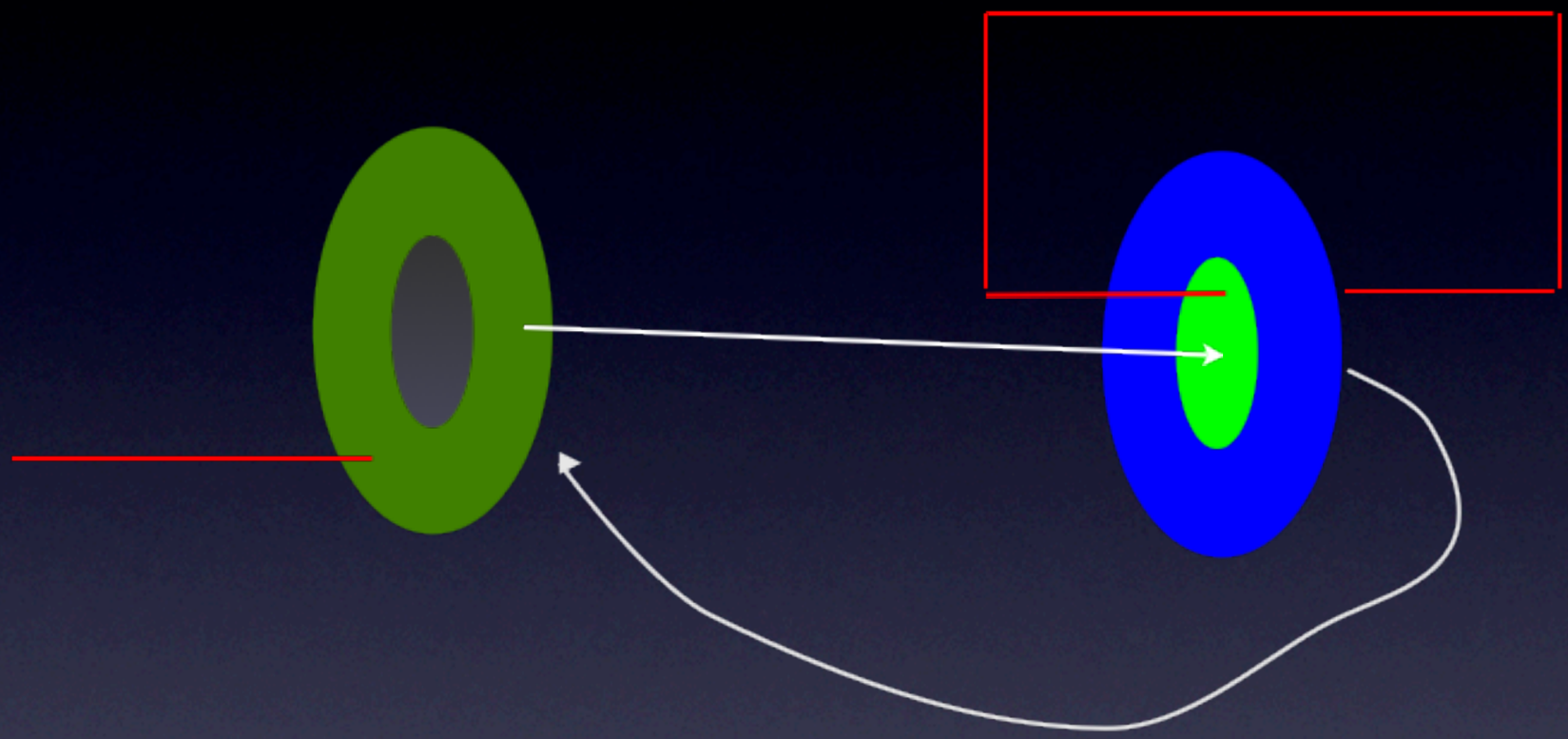
rough edges





two types of qubits

3D Toric Code Defects



performs a controlled not between these qubits

strings can have nontrivial statistics by weaving the strings through each other

Subsystem Codes

Take a subspace $\mathcal{H}_S \subset \mathcal{H}$ and put a tensor product on it

$$\mathcal{H}_S = \mathcal{H}_L \otimes \mathcal{H}_G$$

encode quantum information into $\mathcal{H}_L$

One doesn't care what happens to information in $\mathcal{H}_G$

Quantum information protected up to change in "gauge"

Stabilizer Subsystem Codes

Normalizer of stabilizer in Pauli group:

$$\begin{array}{cccccc} & X_1 & X_2 & \cdots & X_{k'} \\ \mathcal{N}/\mathcal{S} & Y_1 & Y_2 & \cdots & Y_{k'} \\ & Z_1 & Z_2 & \cdots & Z_{k'} \end{array}$$

Divide into gauge and logical qubits:

$$\mathcal{N}/\mathcal{S} = \mathcal{N}_G \otimes \mathcal{N}_L \qquad k' = k + g$$

$$\begin{array}{cccc} X_1 & X_2 & \cdots & X_g \\ Y_1 & Y_2 & \cdots & Y_g \\ Z_1 & Z_2 & \cdots & Z_g \end{array}$$

gauge

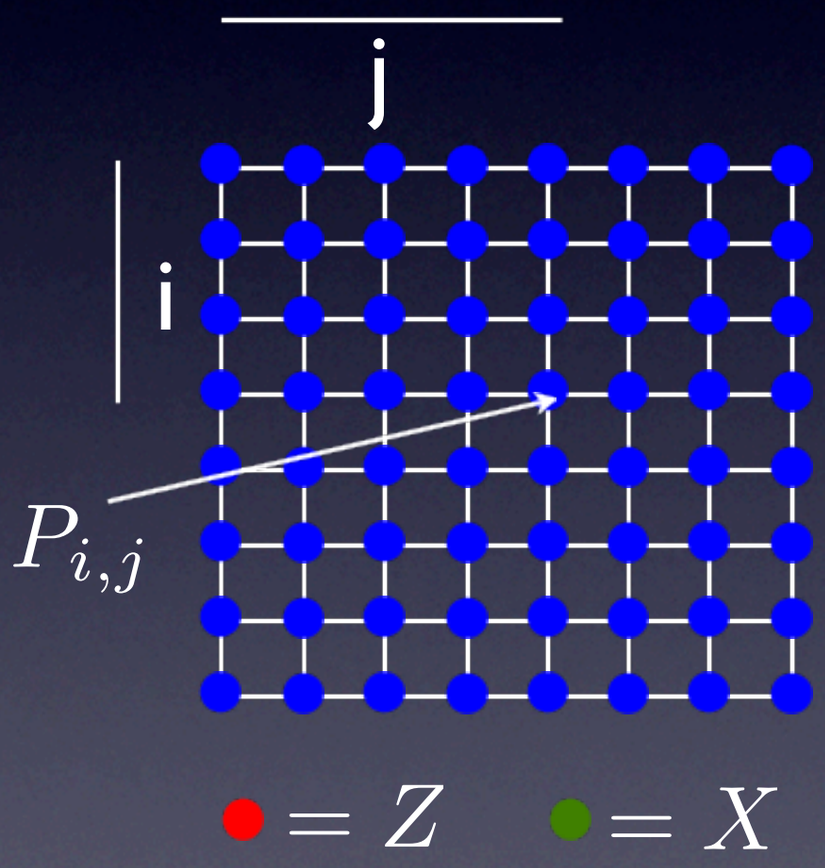
$$\begin{array}{cccc} X_1 & X_2 & \cdots & X_k \\ Y_1 & Y_2 & \cdots & Y_k \\ Z_1 & Z_2 & \cdots & Z_k \end{array}$$

logical

[[n,k,g,d]] subsystem code

Doubling the Toric Code

L x L (L even) square lattice



z type

$$S^z_{i,j} = Z_{i,j} Z_{i+1,j} Z_{i,j+1} Z_{i+1,j+1}$$

x type

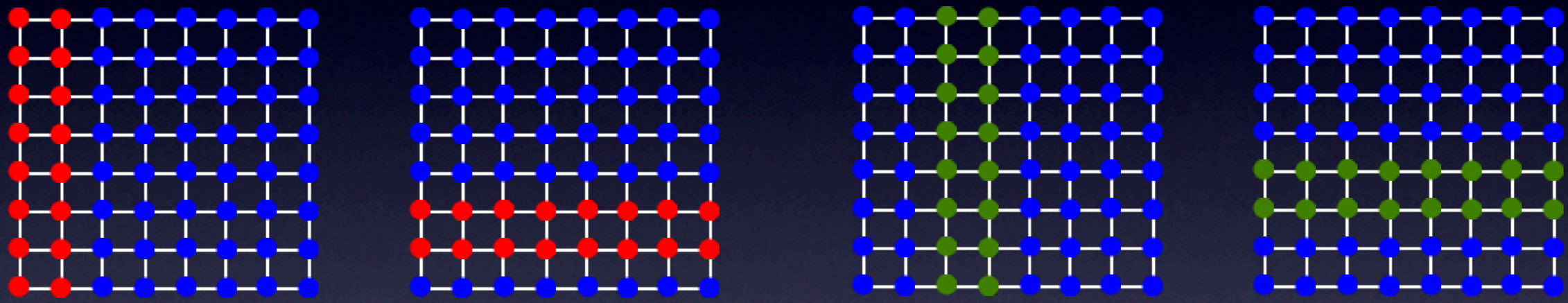
$$S^x_{i,j} = X_{i,j} X_{i+1,j} X_{i,j+1} X_{i+1,j+1}$$

Consider the group generated by these operators:

$$\mathcal{T} = \langle \quad , \quad \rangle$$

Doubled Toric Code

Abelian invariant subgroup:



$$C_j^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i+1,j} \quad C_i^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i,j+1} \quad C_j^X = \prod_{i=1}^{L} X_{i,j} X_{i+1,j} \quad C_i^X = \prod_{i=1}^{L} X_{i,j} X_{i,j+1}$$

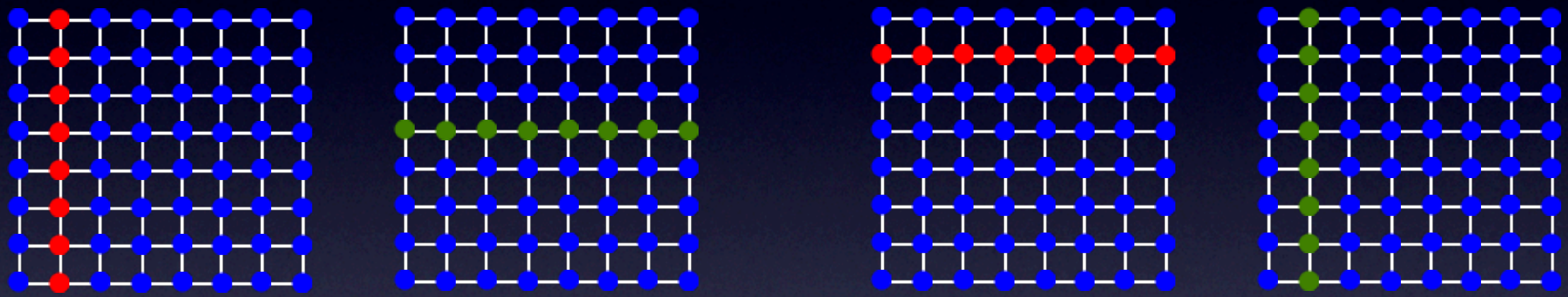
$(L-1)^2 - 1$ generators $(L-1)^2 - 1$ generators

$\bullet = Z \quad \bullet = X$

Stabilizer group

Doubled Toric Code

Normalizer of stabilizer group: $\mathcal{T}$ and two logical qubits



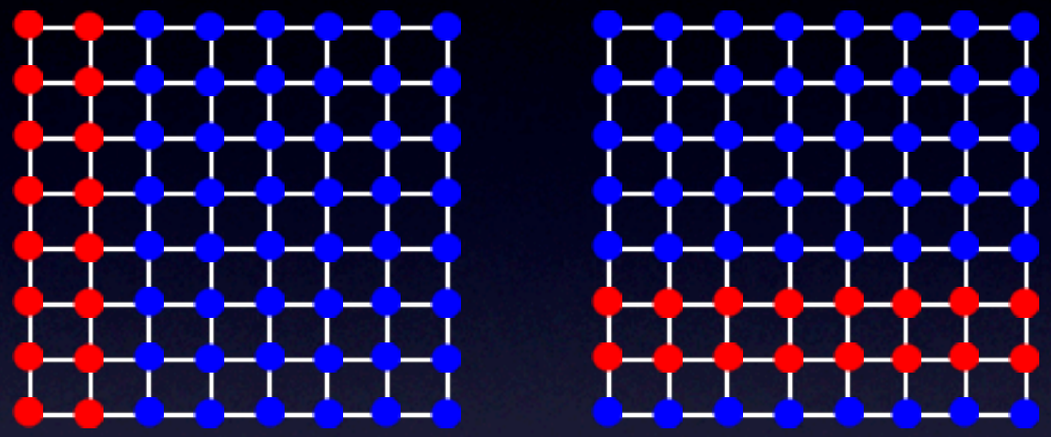
Z_1 X_1 Z_2 X_2

[[L^2,2,L^2-2,L/2]] subsystem code

● $= Z$ ● $= X$

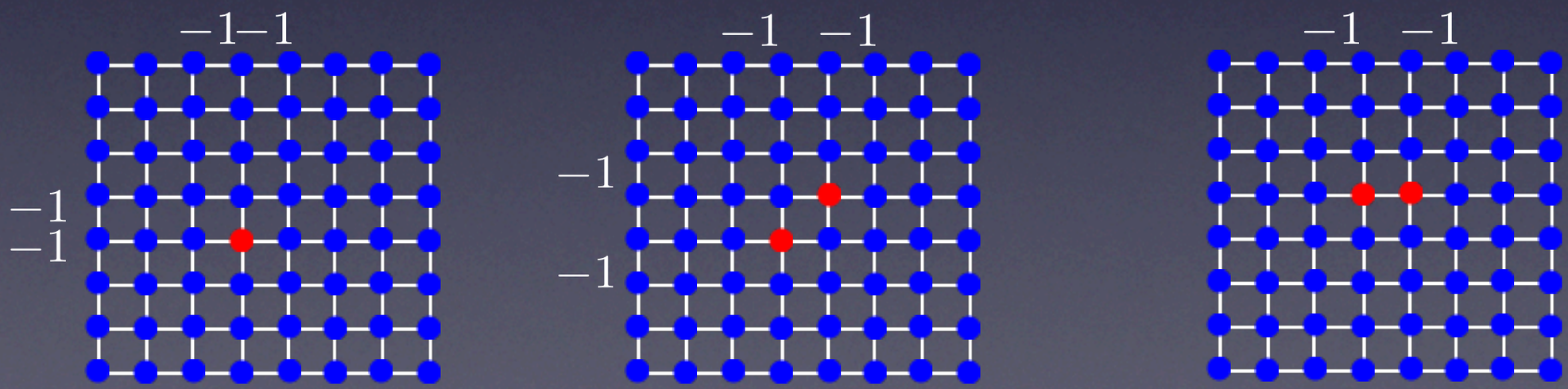
Doubled Toric Code

Error correcting procedure:



treat as redundancy code on boundary

$$C_j^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i+1,j} \qquad C_i^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i,j+1}$$



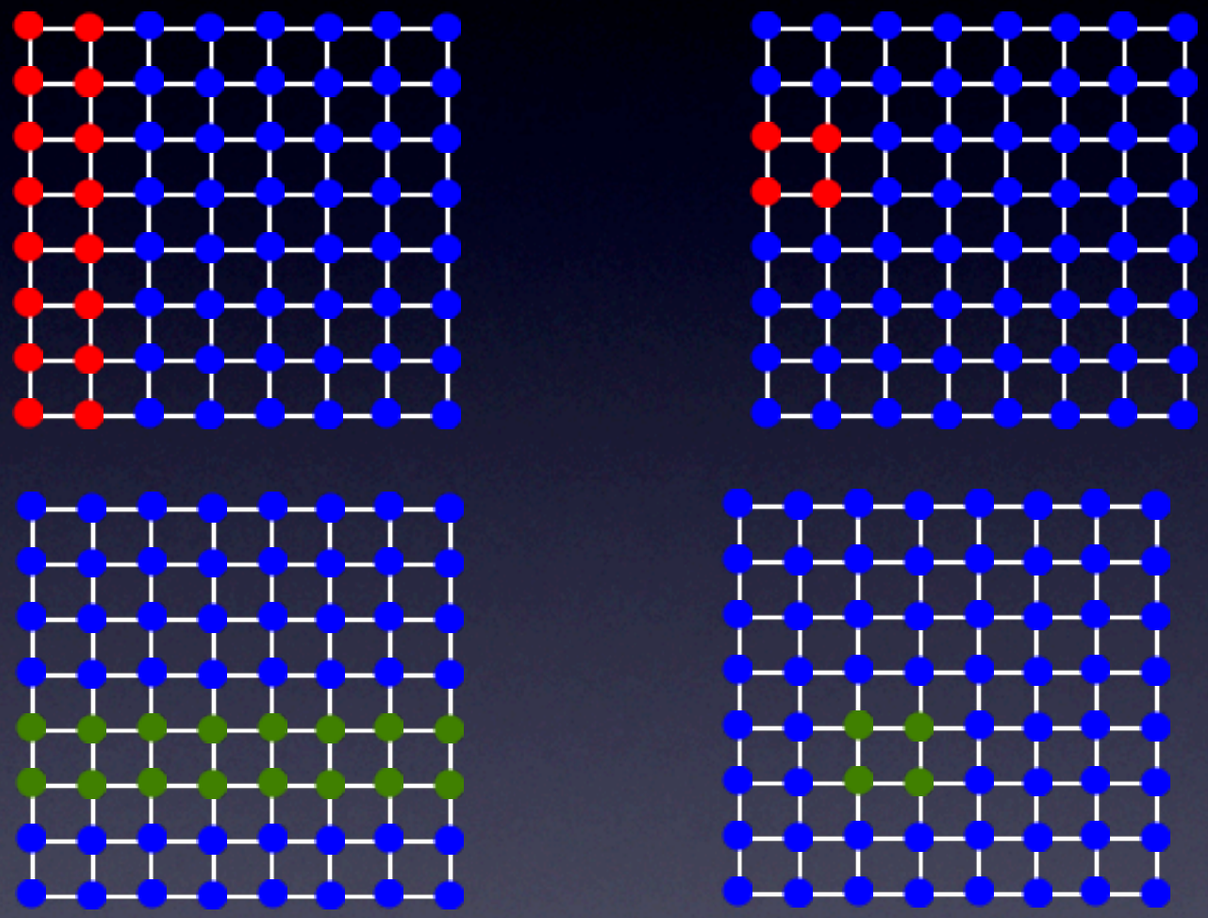
● $= Z$ ● $= X$

matched errors

unmatched

Doubled Toric Code

Fault tolerance and locality

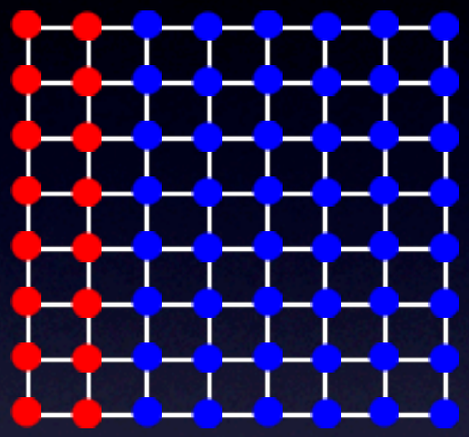


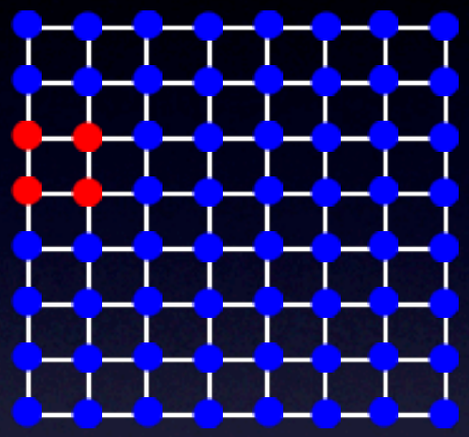
Even though they don't commute we can measure each four qubit operator separately then send info to boundary to correct

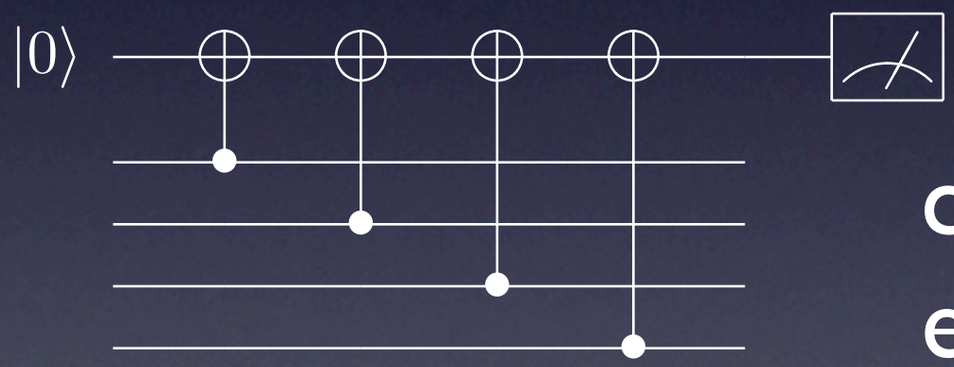
$\bullet = Z \quad \bullet = X$

Doubled Toric Code

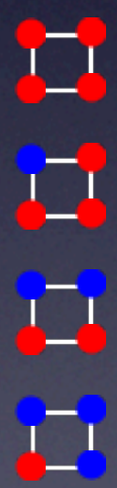
Fault tolerance and locality







can spread Z error to code



gauge

gauge + 1 err

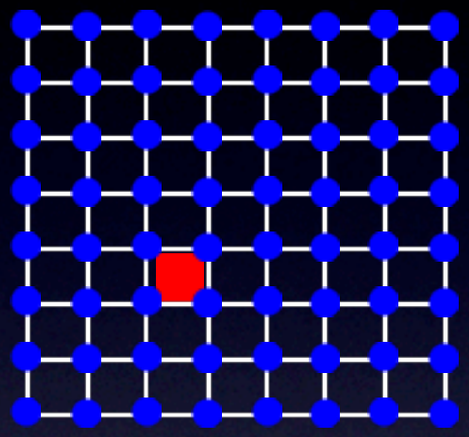
2 err

1 err

If distance of code >5, then 2 errs not a problem

$\bullet = Z \quad \bullet = X$

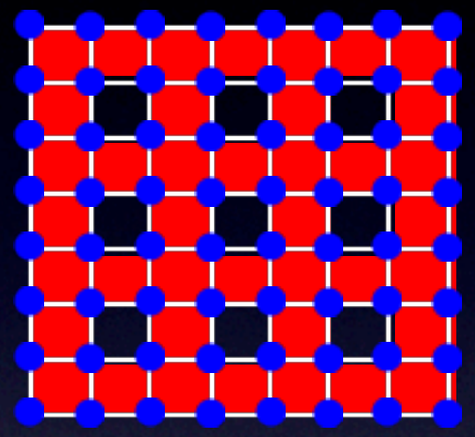
Puncturing the DTC



remove one

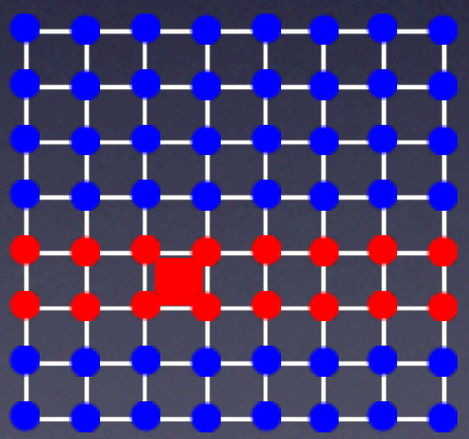


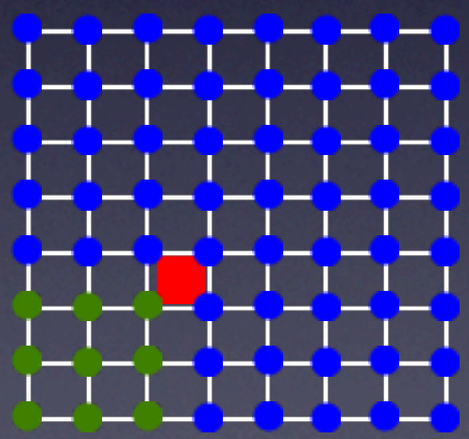
z type



region of allowed removal

Z_L

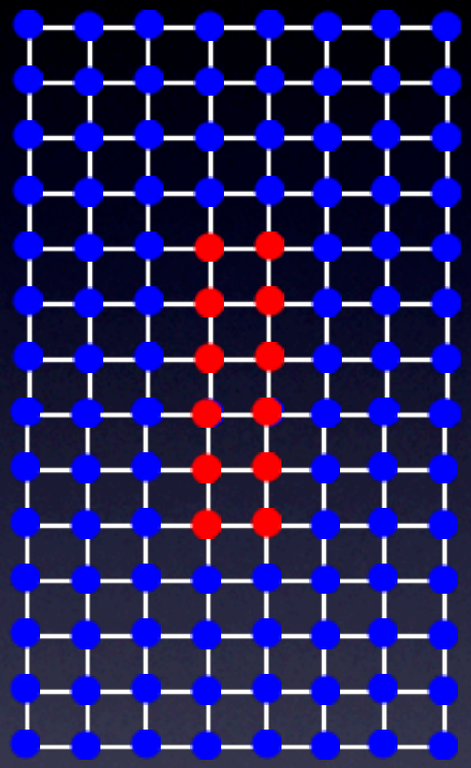




X_L

produces one encoded qubit

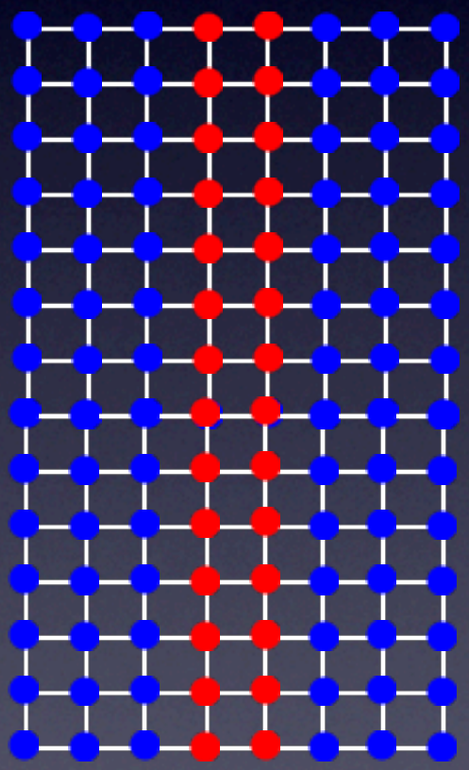
Puncturing the DTC



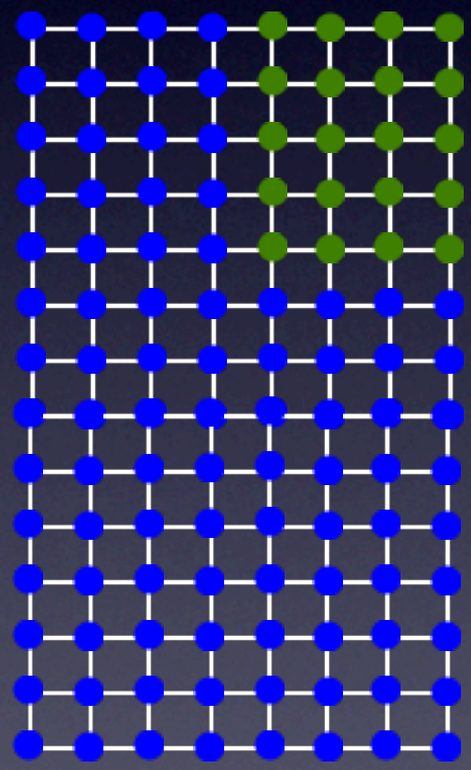
remove > 1?



z type



Z_L

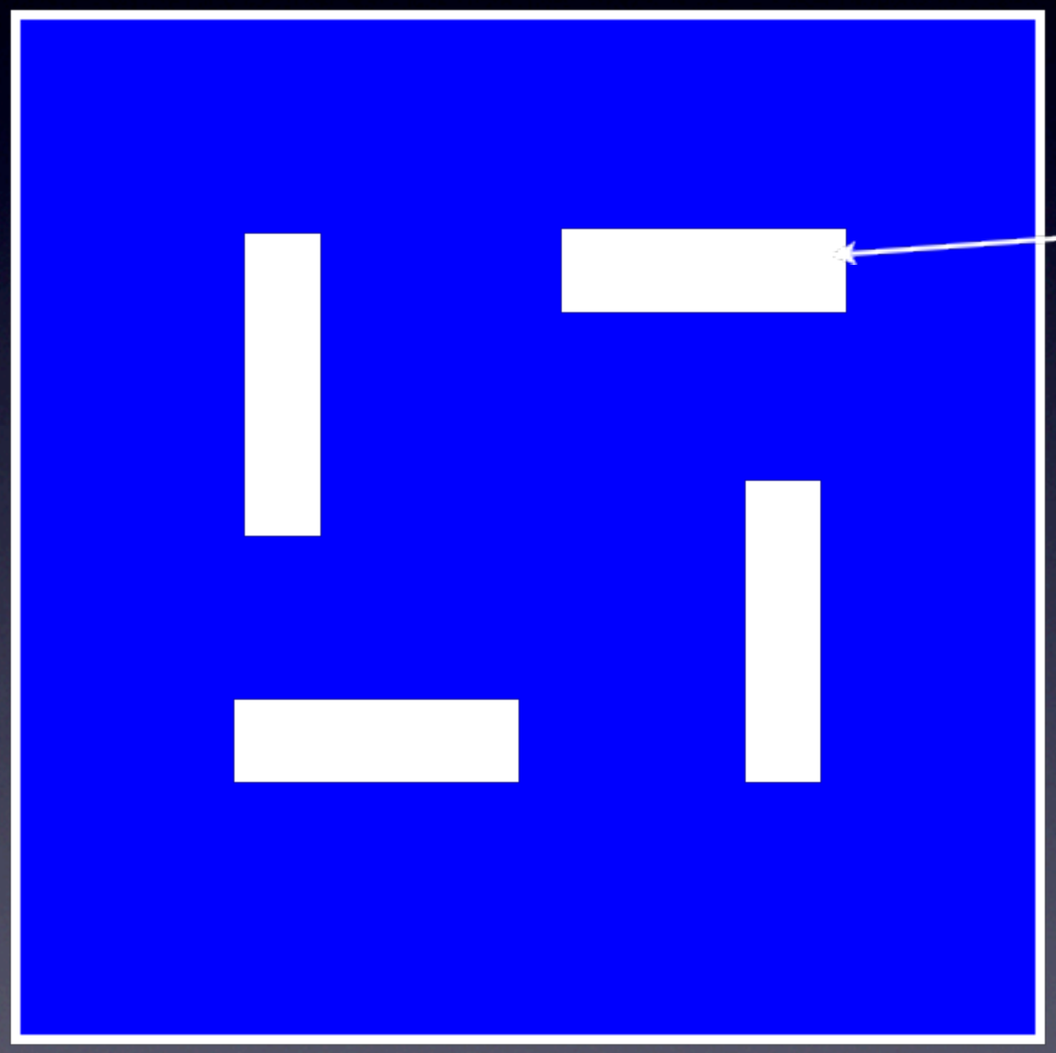


X_L

weight of smallest Z error (mod gauge) = length of defect

Local Architecture?

Build a nearly local quantum error correcting architecture

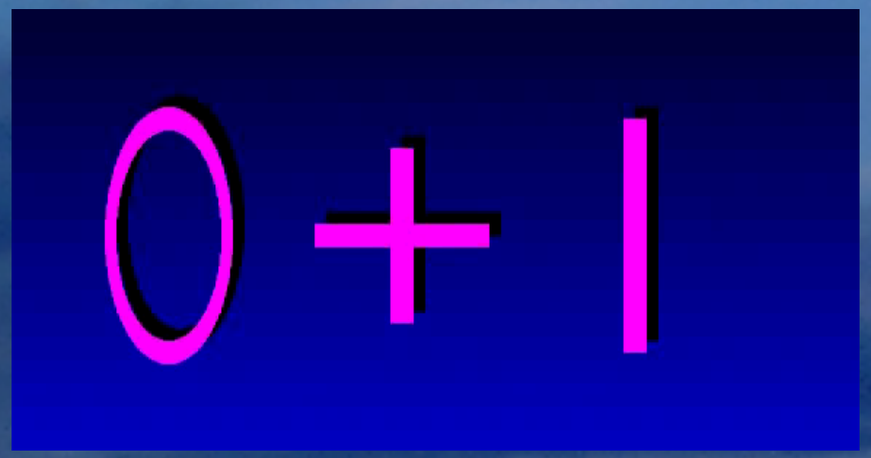


Benefit over toric surface code:
error correction at boundary

Puncturing

Fault-tolerant quantum computing:
physics or computer engineer?

New subsystem code:
fault-tolerant properties
error correction at boundaries
braiding operations?



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